

Introduction

This analysis focuses on predicting the daily stock prices of Automated Data Processing company within the Industrials sector of the S&P 500, using historical data from 11/1/2023 to 10/31/2024. By examining trends, relationships, and influential factors, the study aims to create a reliable model that provides actionable insights into the company's stock behavior and aids in strategic decision-making. The chosen company, Automated Data Processing (ADP), belongs to Human resources and employment contributing significantly to the sector's performance.

The dataset includes daily closing prices for Industrials sector companies, categorized using the Global Industry Classification Standard (GICS). Analytical methods such as Exploratory Data Analysis, correlation analysis, and multiple regression (regression using different selection criterion, stepwise, backward, forward, R-square, adjusted r-square, Cp, press) were applied to identify significant predictors of stock price movements. To refine the model and ensure predictive accuracy, multicollinearity was addressed using stepwise regression, with careful validation of model assumptions to improve reliability and also heteroscedasticity.

Findings revealed a strong relationship between stock prices and several key predictors, including daily prices of Paychex (PAYX), Dayforce (DAY), Broadridge Financial Solutions (BR), Builders FirstSource (BLDR), UPS and United Airlines Holdings (UAL). The final model is a quadratic model and exhibited high explanatory power, as indicated by a strong adjusted R-squared value. Residual diagnostics confirmed the model's validity, showing no major violations of assumptions. The results highlight the influence of sector-specific dynamics and broader market trends on ADP's stock performance, offering valuable insights for stakeholders and investors.

Exploratory Data Analysis

The potential independent variables: Paycom(PAYC), Paychex(PAYX), Dayforce(DAY), Broadridge Financial Solutions(BR), Equifax(EFX), FedEx(FDX), UPS, C.H.Robinson(CHRW), Delta Air Lines(DAL), United Airlines Holdings(UAL), Southwest Airlines(LUV), Carrier Global(CARR), Johnson Controls(JCI), Builders FirstSource(BLDR), Cintas(CTAS), Copart(CPRT), Leidos(LDOS), Fastenal(FAST), United Rentals(URI), Honeywell(HON), 3M(MMM), and Uber(UBER) and the dependent variable, Automatic Data Processing(ADP).

The selection of these companies is based on their strong ties to industries and sub-industries that influence ADP's core operations in payroll, HR, and workforce management. Paycom (PAYC), Paychex (PAYX), and Dayforce (DAY) represent direct competitors in the Human Resource & Employment Services industry, providing key benchmarks for ADP's competitive landscape and insights into market trends such as innovation in HR technologies and customer acquisition strategies. Firms like Broadridge Financial Solutions (BR) and Equifax (EFX), categorized under Data Processing & Outsourced Services and Research & Consulting Services, respectively, play a pivotal role in shaping the demand for advanced data analytics, background checks, and payroll integration—areas critical to ADP's service offerings.

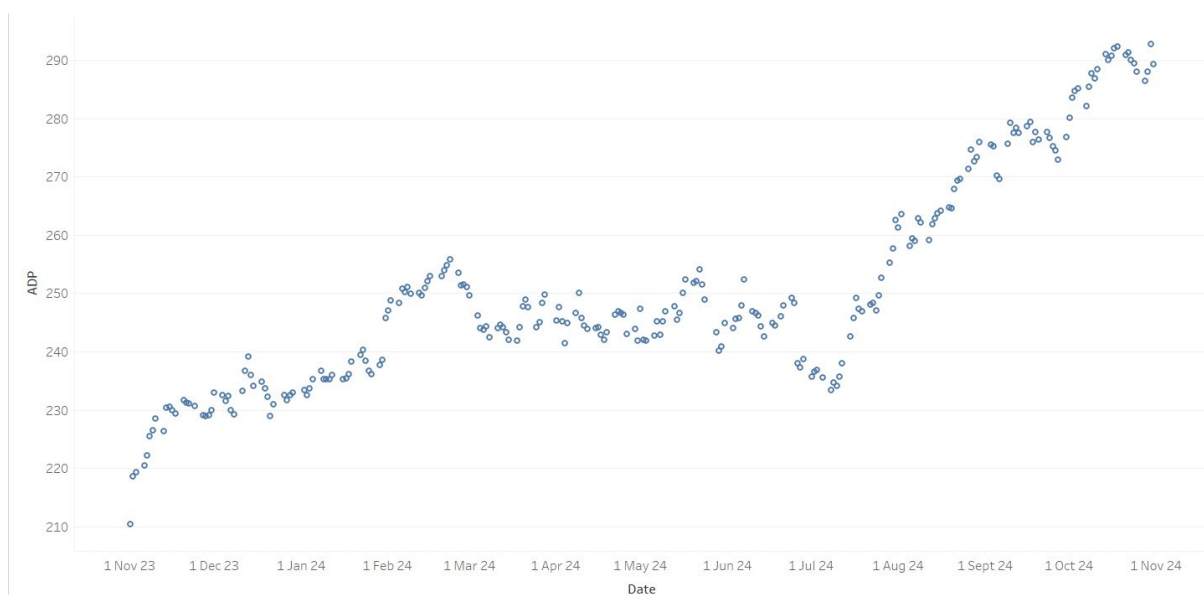
The inclusion of companies from workforce-heavy industries further strengthens the analysis. FedEx (FDX), UPS, and C.H. Robinson (CHRW) from Air Freight & Logistics, as well as airlines like Delta Air Lines (DAL), United Airlines Holdings (UAL), and Southwest Airlines (LUV) from the Passenger Airlines sector, are major employers whose workforce dynamics—expansion, layoffs, or seasonal hiring—are indicative of broader employment trends that impact ADP's client base. Similarly, the Building Products industry, represented

by Carrier Global (CARR), Johnson Controls (JCI), and Builders FirstSource (BLDR), reflects construction and manufacturing employment trends that drive payroll services demand.

Other industries, like Diversified Support Services (Cintas (CTAS), Copart (CPRT), and Leidos (LDOS)), Trading Companies (Fastenal (FAST) and United Rentals (URI)), and Industrial Conglomerates (Honeywell (HON) and 3M (MMM)), provide insights into the demand for outsourced HR and workforce solutions across diverse sectors. Additionally, the inclusion of Uber (UBER) highlights ADP's potential engagement with gig economy platforms, which require tailored payroll solutions. By analysing companies across these interconnected industries, we capture the multifaceted factors that drive ADP's business performance, ensuring a well-rounded and comprehensive approach to understanding its stock behavior.

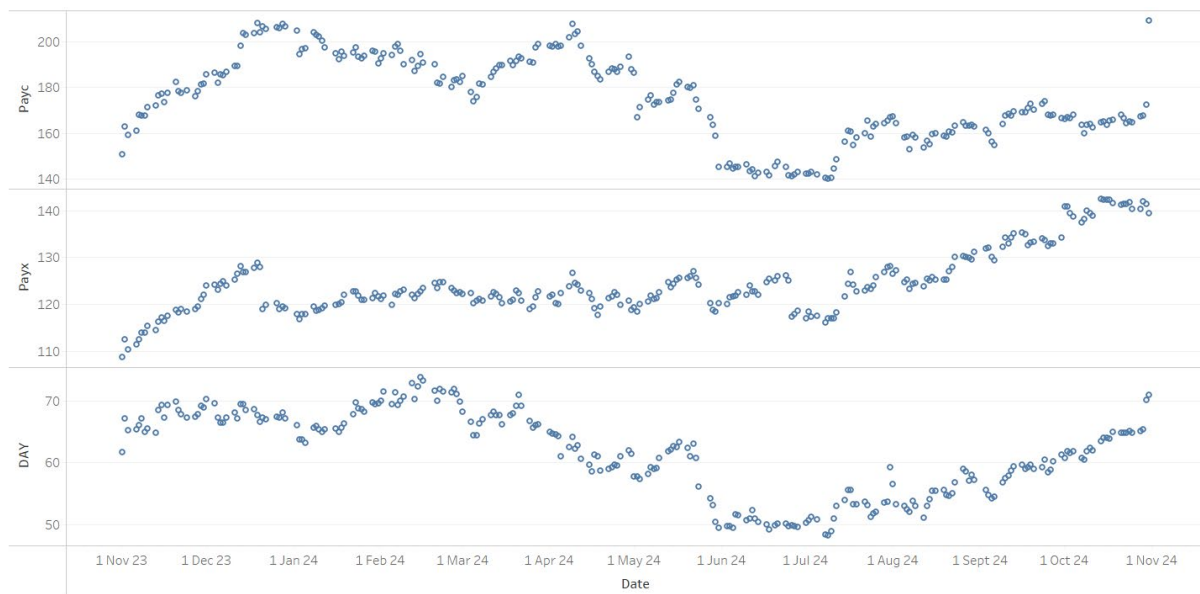
Scatterplots of the companies corresponding to the month:

For ADP:



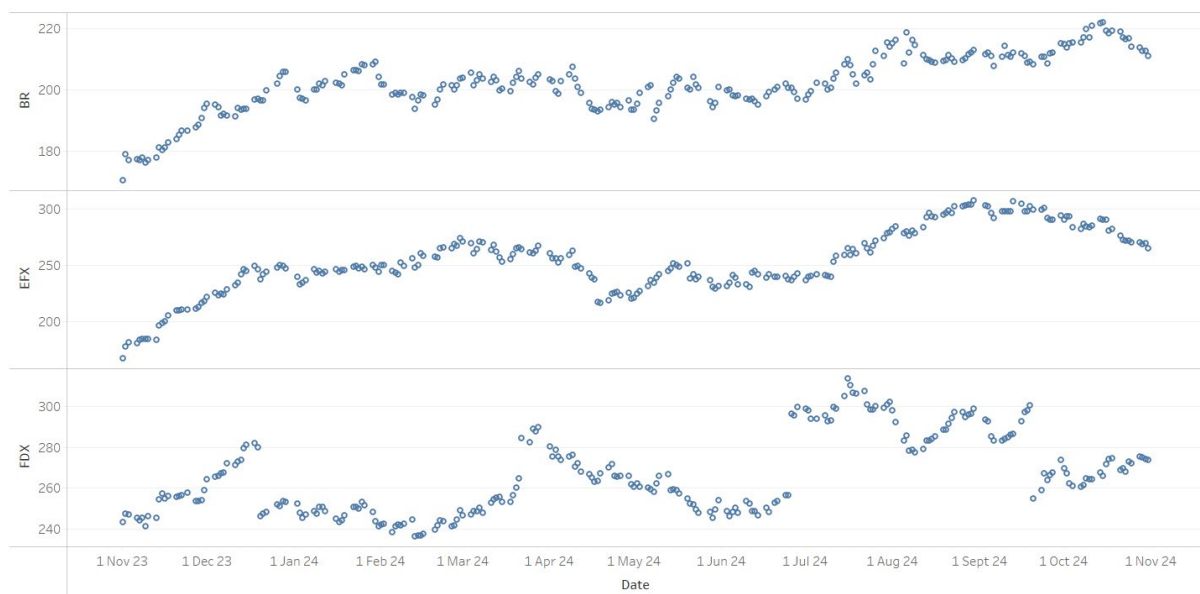
There are ups and downs in the daily prices of ADP, but after a year there is a noticeable increase. No potential outliers.

For PAYC, PAYX, DAY:



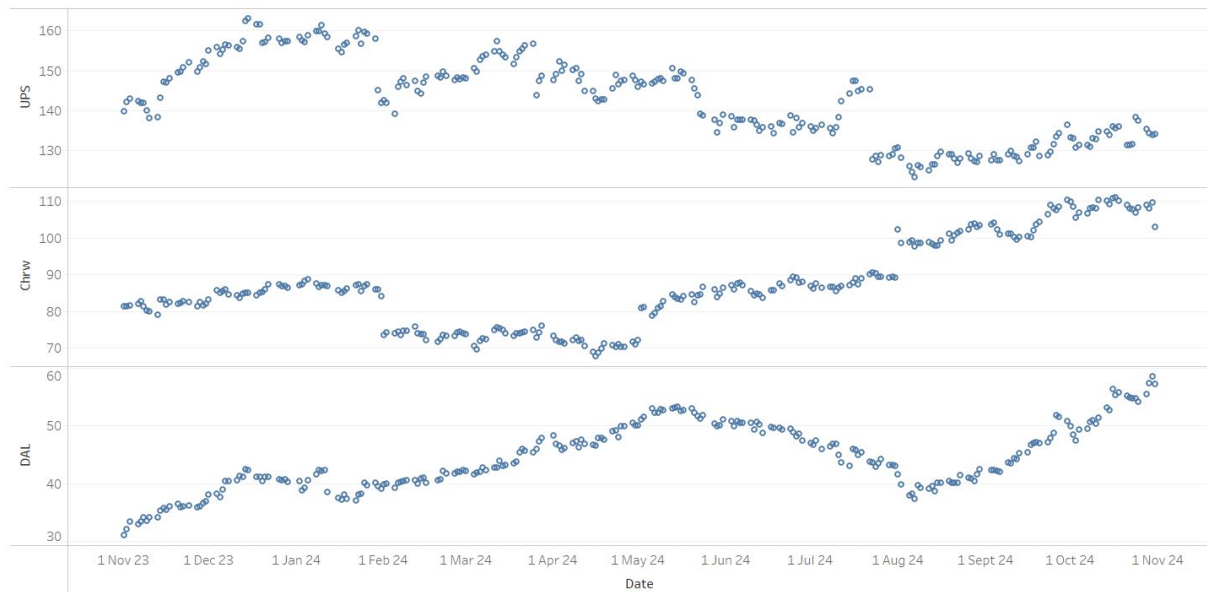
For PAYX, there is a noticeable increase in daily prices over the year but for PAYC and DAY, there doesn't seem to be much increase. One potential outlier for PAYC.

For BR, EFX and FDX:



There is a noticeable increase in daily prices for BR, EFX and FDX over the year but there are some irregularities in the graph for FDX. No potential outliers.

For UPS, CHRW and DAL:



There is a significant increase in daily prices for DAL and CHRW and a noticeable decrease in daily prices for UPS over the year. No potential outliers.

For UAL, CHRW and LUV:



There has been a significant increase in the daily prices for UAL, CHRW and LUV over the year. There are some irregularities in the graph for LUV. No potential outliers.

For JCI, BLDR and CTAS:



There has been a noticeable increase in daily prices for JCI, BLDR and CTAS over the year.

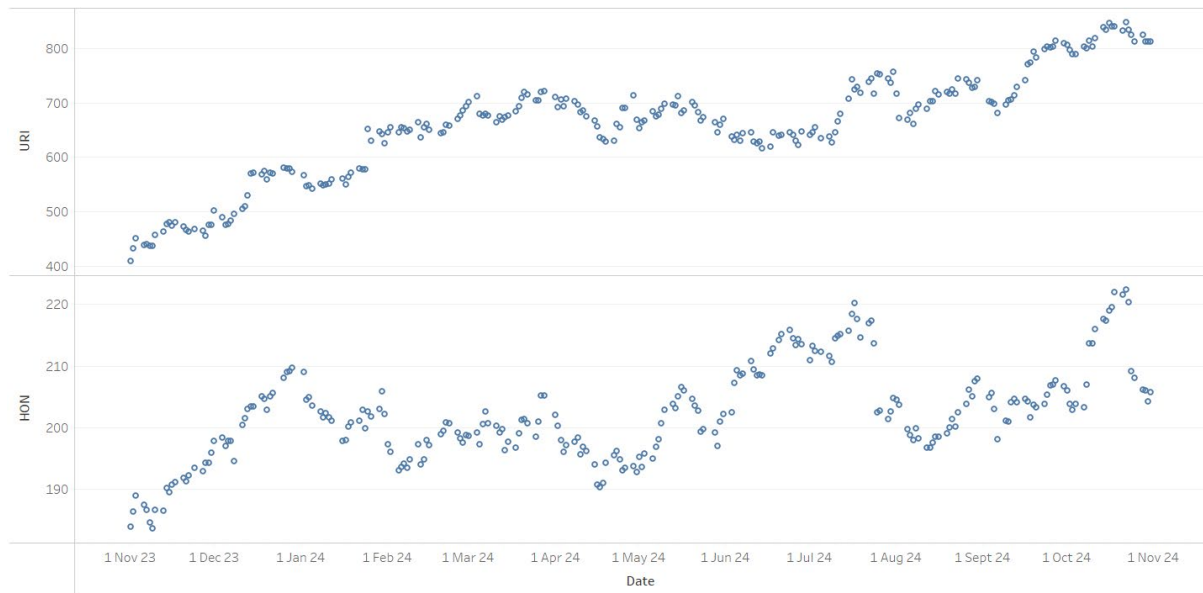
No potential outliers.

For FAST, LDOS and CPRT:



There is a good difference in daily prices of FAST and LDOS but a bit less for CPRT over the year. No potential outliers.

For HON and URI:



There is significant increase in daily prices of URI and a bit less for HON. There are many ups and downs in the daily prices of HON over the year. No potential outliers.

For MMM and UBER:



There has been a significant increase in the daily prices of UBER and MMM over the year.

One potential outlier for UBER.

Methods

The analysis was conducted using various SAS procedures to examine relationships and predict team wins. For correlation analysis, the PROC CORR procedure was employed to quantify associations between variables. To visually assess relationships between metrics, the PROC SGPLOT procedure generated scatter plots, providing a clear view of patterns and potential trends. Interaction effects were evaluated using both PROC GLM and PROC PLM, enabling the exploration of how variables might jointly impact daily prices of ADP. PROC GLMSELECT was used for press criterion and BIC criterion. Finally, PROC REG and selection was used to perform multiple regression analysis, identifying the best combination of predictors with statistical significance. These procedures together allowed for a comprehensive approach to analysing and modeling the data.

Correlation Analysis

We will now conduct a correlation analysis to identify significant relationships between the independent variables and the dependent variable. The goal is to determine whether any correlations are strong enough to justify using the independent variables as predictors in a regression analysis. Pearson's correlation coefficient (r) will be used to measure these associations.

I will now assess the relationships between each one of the potential independent variables: Paycom(PAYC), Paychex(PAYX), Dayforce(DAY), Broadridge Financial Solutions(BR), Equifax(EFX), FedEx(FDX), UPS, C.H.Robinson(CHRW), Delta Air Lines(DAL), United Airlines Holdings(UAL), Southwest Airlines(LUV), Carrier Global(CARR), Johnson Controls(JCI), Builders FirstSource(BLDR), Cintas(CTAS), Copart(CPRT), Leidos(LDOS), Fastenal(FAST), United Rentals(URI), Honeywell(HON), 3M(MMM) and Uber(UBER) and the dependent variable, Automatic Data Processing(ADP). The significance of the correlation can be evaluated by examining the p-value. If the p-value is less than the alpha level (0.05), we will reject the null hypothesis—which indicates that there is no correlation between the two variables. Conversely, if the p-value exceeds 0.05, we will fail to reject the null hypothesis, indicating that the correlation is not significant. If a significant correlation is found, we will then evaluate the strength of the relationship using the r value.

The correlation coefficient ranges from -1 to 1. An r of -1 or 1 indicates a perfect correlation. An r value between 0.7 and 0.9 or between -0.7 and -0.9 signifies a strong relationship, while an r -value between 0.4 and 0.7 or between -0.4 and -0.7 indicates a moderately strong relationship. An r between 0.1 and 0.4 or between -0.1 and -0.4 reflects a weak relationship, and if r is close to 0, it suggests little to no linear relationship. A positive

sign indicates that as one variable increases, the other also increases, whereas a negative sign suggests that as one variable increases, the other decreases.

Here is the table with correlation coefficients for the independent variables with the dependent variable (ADP):

Independent Variable	Correlation Coefficient	p-value
PAYC	-0.26741	<.0001
PAYX	0.92072	<.0001
DAY	-0.15557	0.0134
BR	0.81515	<.0001
EFX	0.81480	<.0001
FDX	0.33542	<.0001
UPS	-0.65186	<.0001
CHRW	0.73317	<.0001
DAL	0.52445	<.0001
UAL	0.69047	<.0001
LUV	0.33164	<.0001
CARR	0.87397	<.0001
JCI	0.79221	<.0001
BLDR	0.50218	<.0001
CTAS	0.89828	<.0001
CPRT	0.24942	<.0001
LDOS	0.82383	<.0001
FAST	0.59671	<.0001
URI	0.85759	<.0001

HON	0.42915	<.0001
MMM	0.84692	<.0001
UBER	0.64009	<.0001

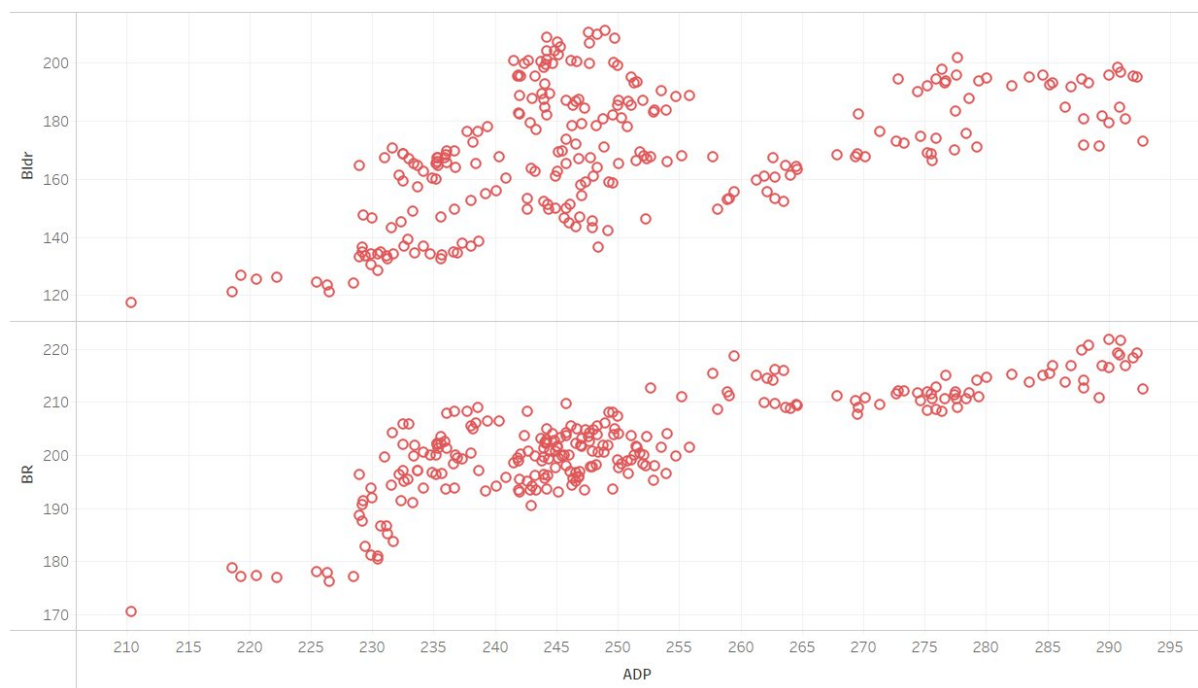
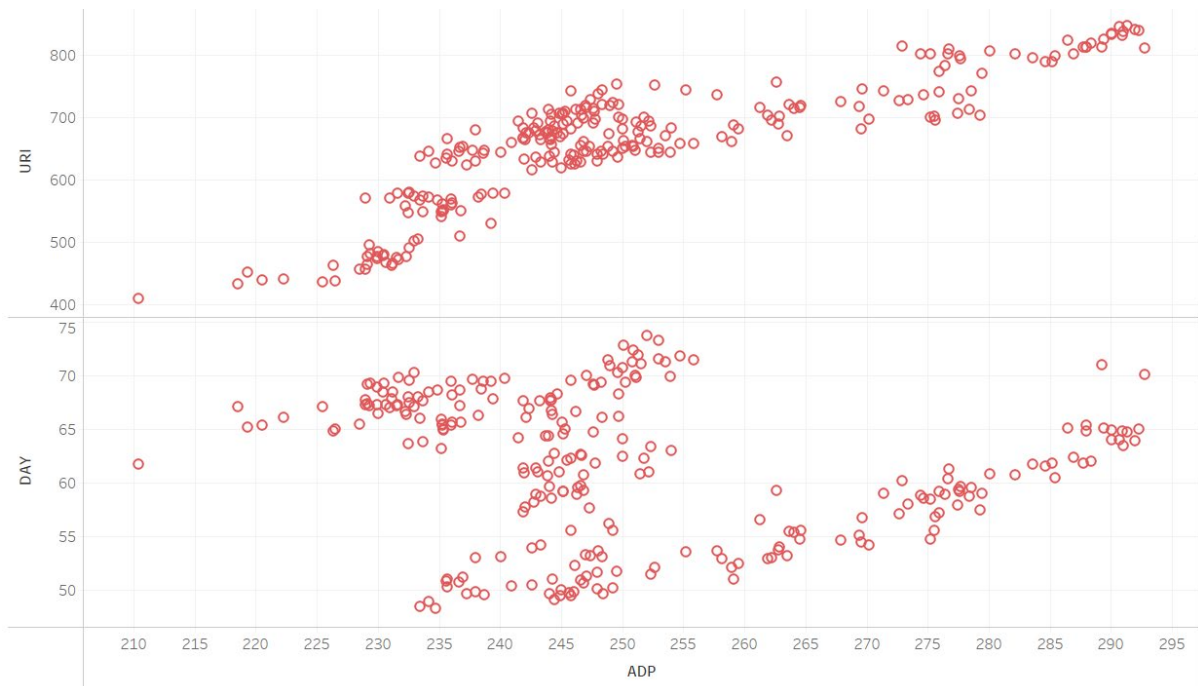
All correlations show significant p-values ($p < 0.05$), allowing us to reject the null hypothesis and conclude that there is a relationship between the chosen independent variables and ADP.

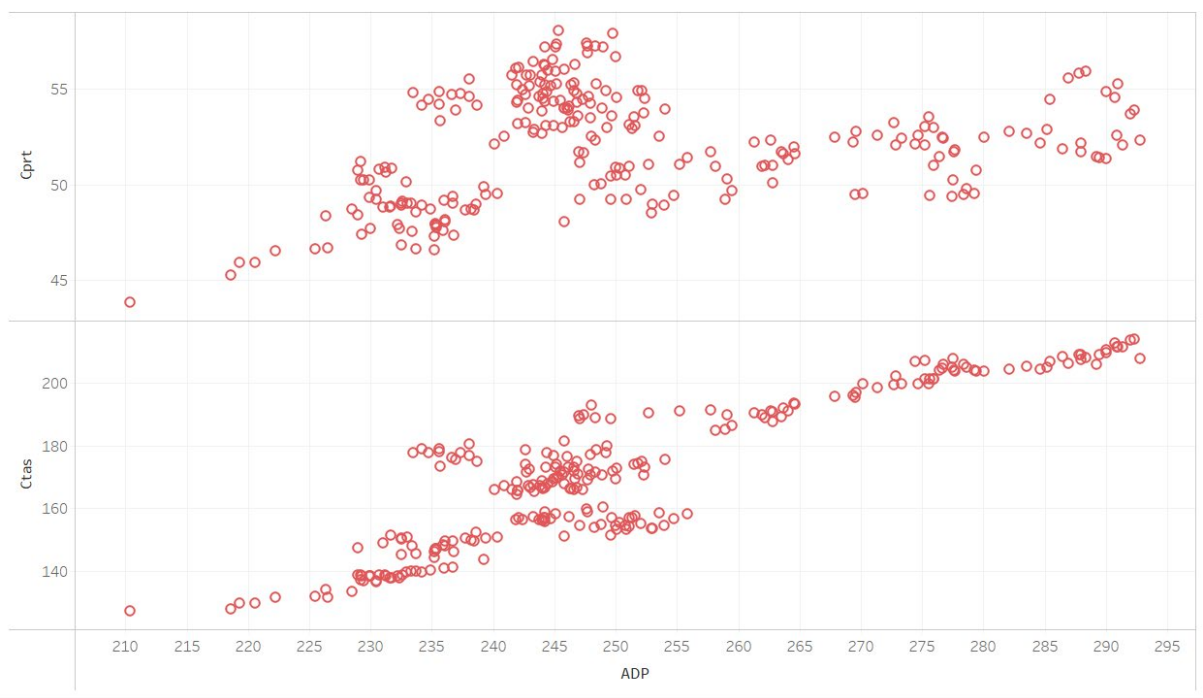
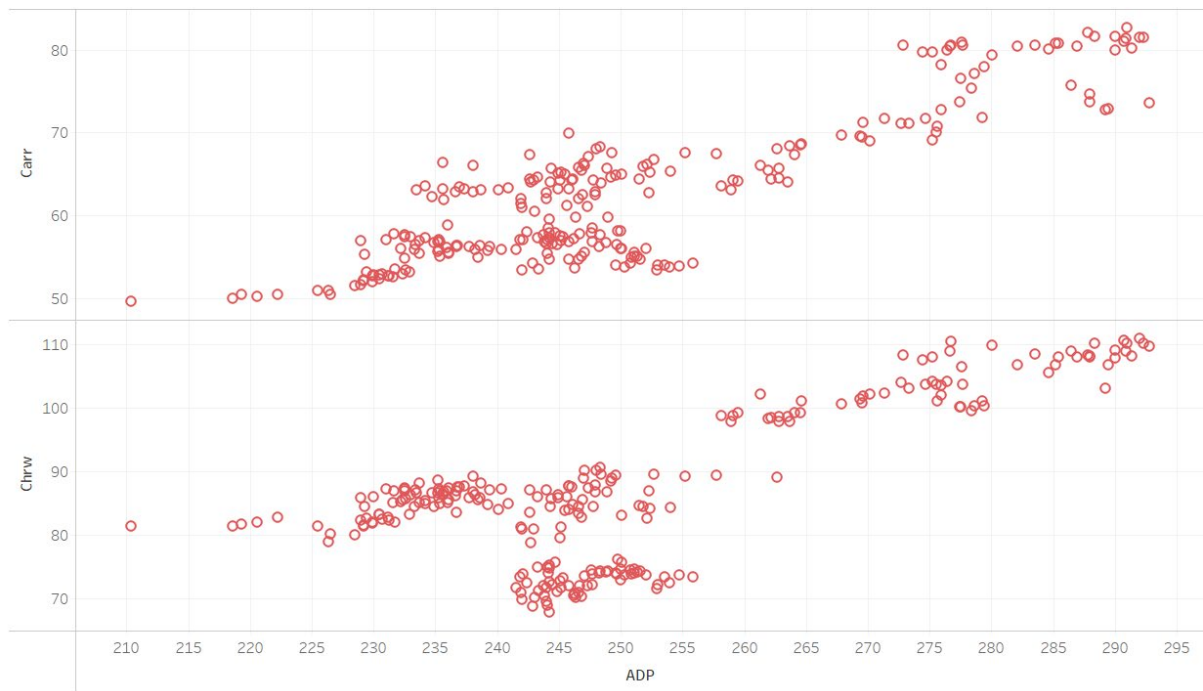
The strength of these relationships can be determined from the r values: there is a strong relationship between PAYX and ADP ($r = 0.92072$), BR and ADP ($r = 0.81515$), EFX and ADP ($r = 0.81480$), CHRW and ADP ($r = 0.73317$), CARR and ADP ($r = 0.87397$), JCI and ADP ($r = 0.79221$), CTAS and ADP ($r = 0.89828$), LDOS and ADP ($r = 0.82383$), URI and ADP ($r = 0.85759$) and, MMM and ADP ($r = 0.84692$).

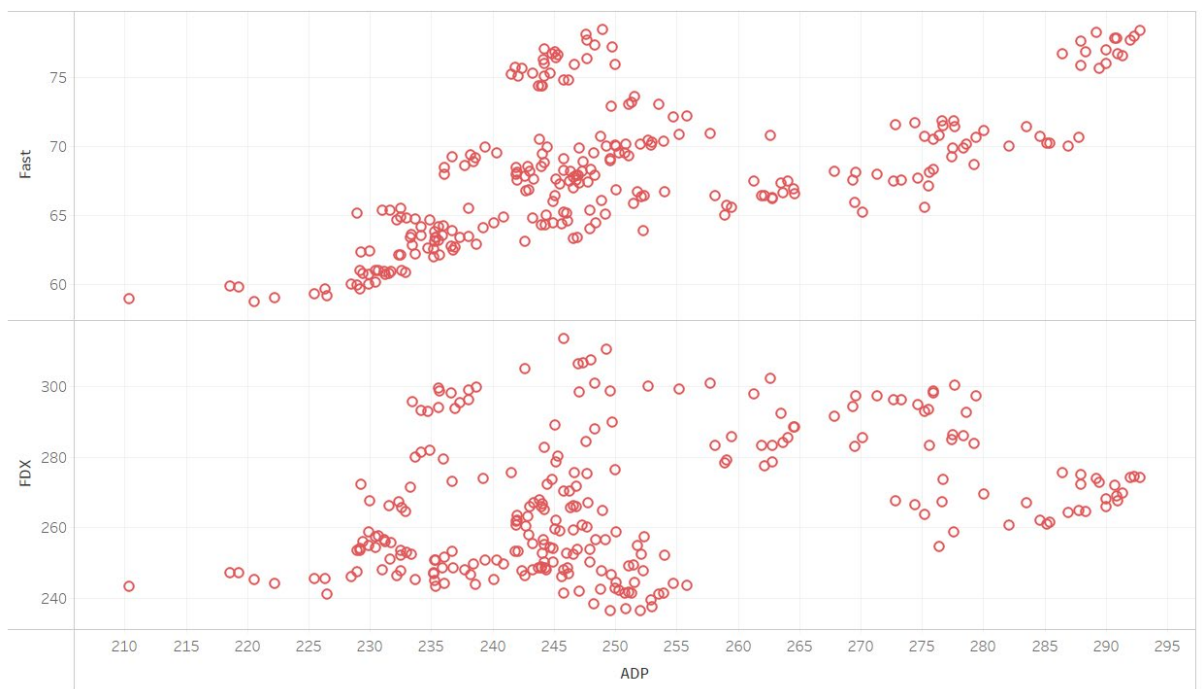
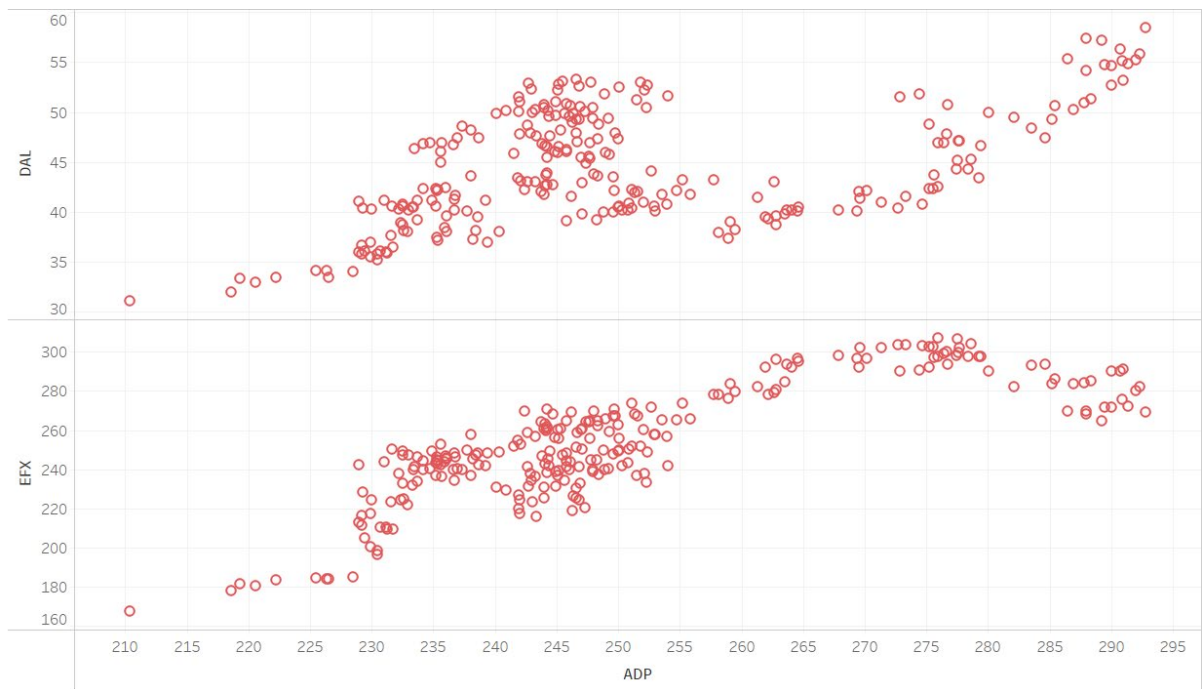
A moderate relationship exists between UPS and ADP ($r = -0.65186$), DAL and ADP ($r = 0.52445$), UAL and ADP ($r = 0.69047$), BLDR and ADP ($r = 0.50218$), FAST and ADP ($r = 0.59671$), HON and ADP ($r = 0.42915$) and UBER and ADP ($r = 0.64009$).

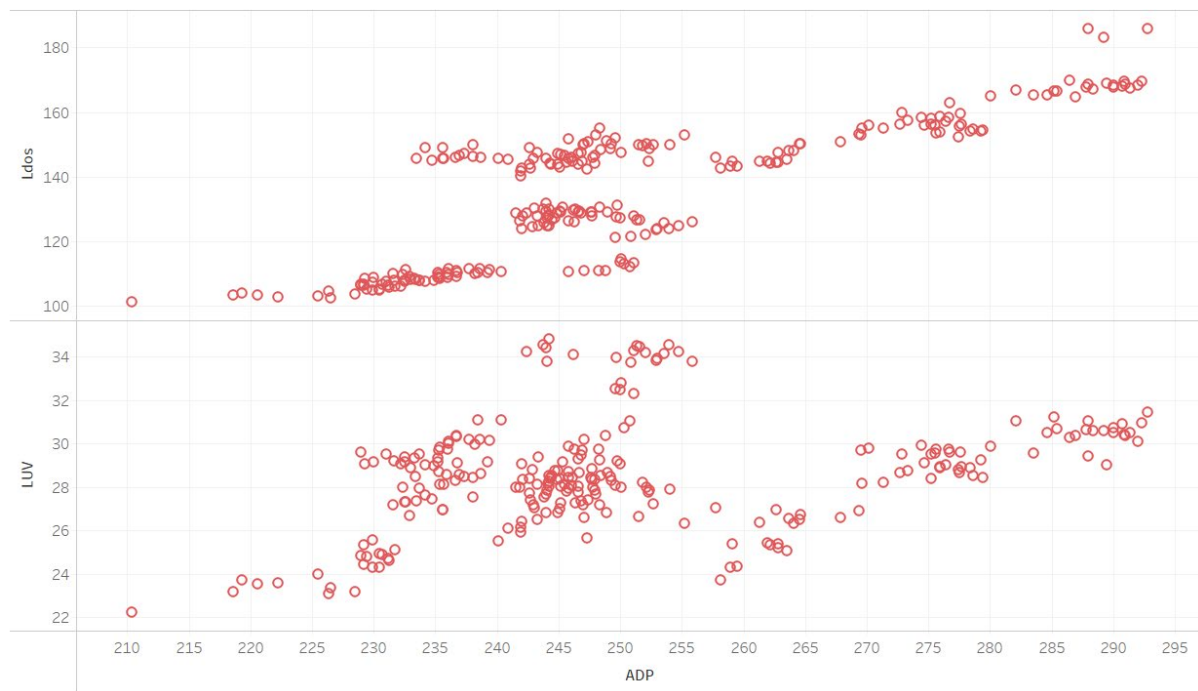
There is a weak relationship between PAYC and ADP ($r = -0.26741$), DAY and ADP ($r = -0.15557$), FDX and ADP ($r = 0.33542$), LUV and ADP ($r = 0.33164$) and CPRT and ADP ($r = 0.24942$).

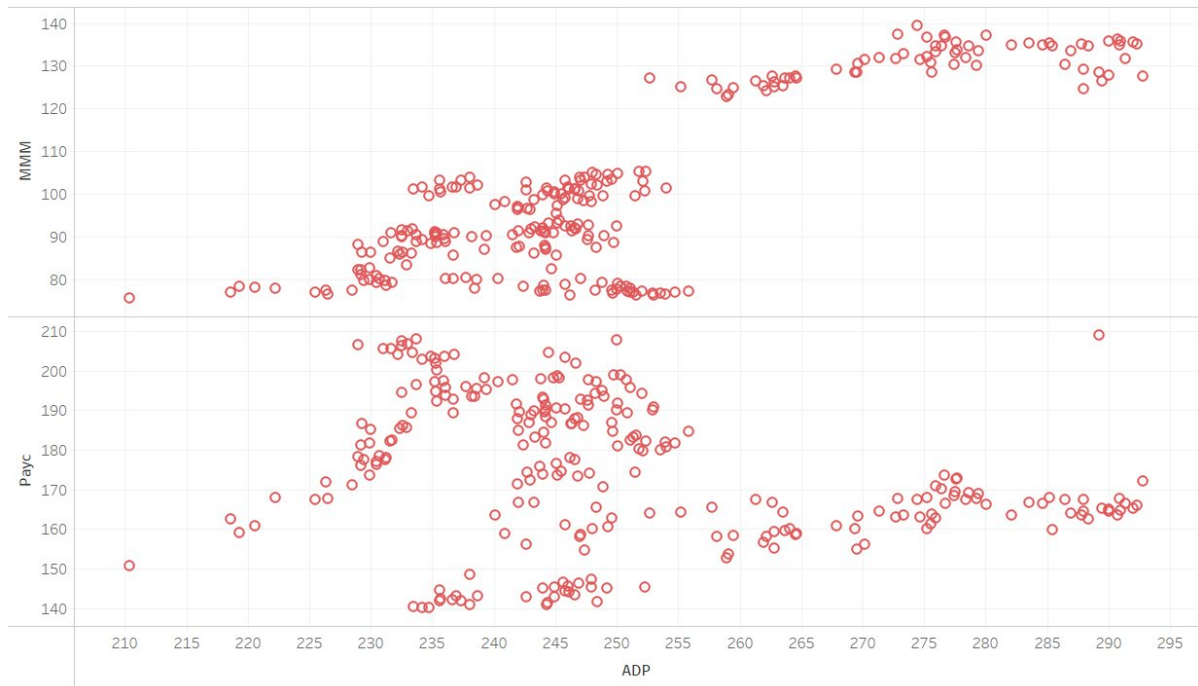
Here are the scatter plots:

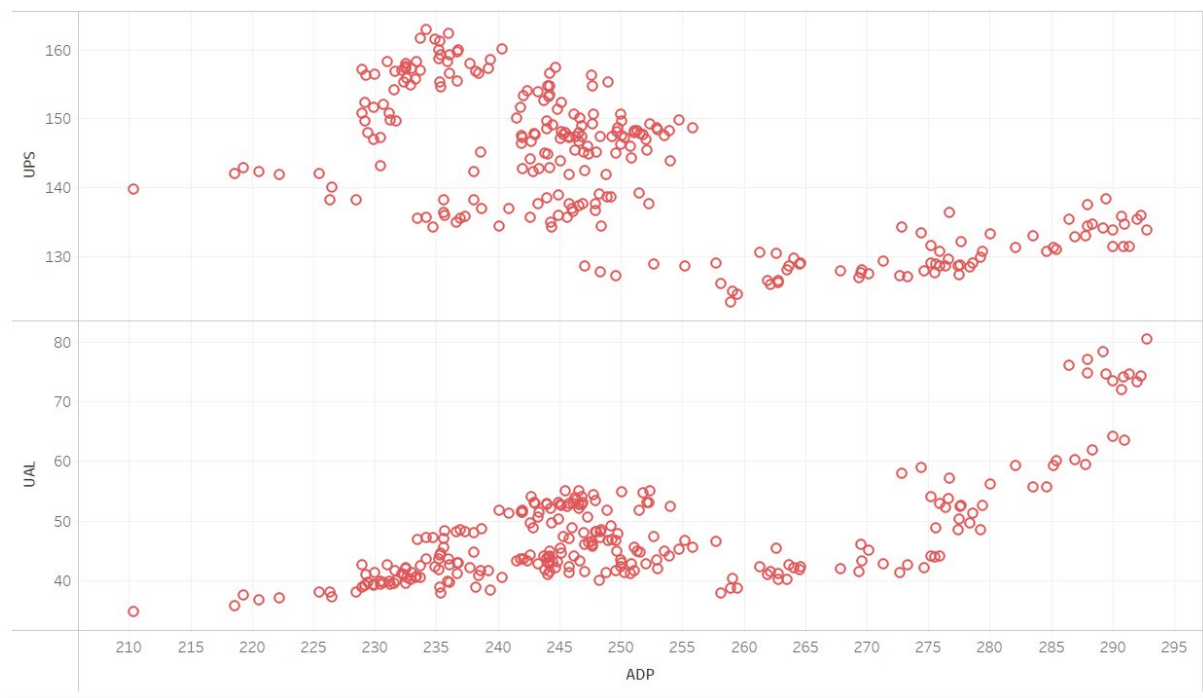












Multiple Regression

First, we need to conduct a multicollinearity test. If two or more of the independent variables used in the model for $E(y)$ contribute redundant information, they will be correlated with each other causing multicollinearity. If the variance inflation is greater than 10, then it is a sign of multicollinearity. First conduct the multicollinearity test for all the variables and remove the highest one first (only if it's above 10), then do the test again, once again remove the highest one, do the test again. Repeat this process until there are no variables with variance inflation greater than 10.

By conducting this test for the 22 variables, we can remove, EFX, CHRW, DAL, CARR, JCI, CTAS, LDOS, URI, MMM and UBER.

In datasets with a large number of variables, each serving as a potential predictor, including too many variables in a model can lead to several issues. Models with numerous predictors can become complex and difficult to manage, making the interpretation of β -coefficients challenging or even impossible. Additionally, such models often require very large sample sizes to provide reliable estimates. In our case, we started with 22 independent variables, which can lead to overfitting and complicate the model. To address these challenges, a process called variable screening is used to identify a subset of variables that contribute to a meaningful and practical model.

Stepwise regression is one such variable screening method, where the model iteratively selects predictors based on their statistical significance. The outcome is a streamlined model containing only predictors with significant t-values. For predicting the daily prices of ADP, stepwise regression with an α -value of 0.05 was applied. This process identified the potential independent variables: PAYX, UPS, BLDR, DAY, BR, UAL, FAST,

PAYC, HON as shown in Figure 1 in appendix. The model obtained with these variables is model 1.

It is important to note that the results of the stepwise procedure do not guarantee the identification of all relevant independent variables or the elimination of all irrelevant ones. Since stepwise regression relies on sample estimates to select variables, there is a high likelihood of Type I Errors (including unimportant variables) or Type II Errors (excluding important variables). Additionally, the procedure may not account for higher-order terms or interactions that could be significant in explaining the dependent variable. Therefore, while stepwise regression serves as the primary variable selection technique, it is important to complement it with other selection criteria methods, such as forward selection, backward elimination, all possible regression selection procedures (with 4 criteria to select from) to ensure a more robust model and support the conclusions drawn from the analysis.

Forward selection puts variables in one at a time but doesn't go back and recheck the t-values that have entered the model. This process identified the potential independent variables: PAYX, UPS, BLDR, DAY, BR, UAL, FAST, PAYC, HON as shown in Figure 2. The model obtained with these variables is model 2.

Backward elimination starts with all variables and then start eliminating based on the smallest t-statistic. This process identified the potential independent variables: PAYC, PAYX, DAY, BR, UPS, UAL, BLDR, FAST, HON. Variables CPRT, FDX, and LUV are removed, as shown in Figure 3. The model obtained with these variables is model 3.

All possible regression selection procedure with R^2 criterion's objective is to find a subset model so that adding more variables to the model will only yield small increases in R^2 . This process identified 2 sets of independent variables with the highest and same R^2 (0.9694).

They are: One, PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON (Model 4) and two, PAYC PAYX DAY BR FDX UPS UAL LUV BLDR CPRT FAST HON (Model 5).

All possible regression selection procedure with Adjusted R^2 or MSE criterion, since R^2 doesn't account for the number of β parameters in the model, adjusted R^2 will only increase if MSE decreases and cannot force to $R^2 = 1$. This allows us to search for the model with the minimum MSE. This process identified 2 sets of independent variables with the highest and same adjusted R^2 (0.9680). They are: One, PAYC PAYX DAY BR UPS UAL LUV BLDR FAST HON (model 6) and two, PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON (model 7).

All possible regression selection procedure with C_p criterion focuses on minimizing the total mean square error and the regression bias. It selects the model with a small value of C_p and a C_p value of near $p+1$ (slight or no bias), where p is the number of independent variables. Using this process, PAYC PAYX DAY BR UPS UAL LUV BLDR FAST HON (model 8), PAYC PAYX DAY BR UPS UAL BLDR FAST HON (model 9) and PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON (model 10) are selected, as their values are 9.5257, 10.1194 and 11.0913 respectively.

All possible regression selection procedure with press criterion (prediction sum of squares), the objective is to fit to the sample data n times, each time omitting one of the data points and obtaining the predicted value of y for that data point. Since small differences indicate the model is predicting well, the goal is to find the model with the smallest press value. This process identified the potential independent variables: PAYX DAY BR UPS UAL BLDR (model 11).

We can also use SBC model selection to get the variables. The Schwarz Bayesian Criterion (SBC), also known as the Bayesian Information Criterion (BIC), is a tool used in model selection to find the best balance between how well a model fits the data and how complex it is. It adds a penalty for including too many predictors, helping to avoid overfitting. The goal is to choose the model with the lowest SBC value, which indicates the simplest model that still explains the data effectively. SBC is stricter than similar criteria like AIC, especially for larger datasets, as it favors simpler models more strongly. Using this process, we get PAYX, UPS, BLDR, DAY, BR and UAL as our potential variables.

Model	variables
Model 1	PAYX, UPS, BLDR, DAY, BR, UAL, FAST, PAYC, HON
Model 2	PAYX, UPS, BLDR, DAY, BR, UAL, FAST, PAYC, HON
Model 3	PAYC, PAYX, DAY, BR, UPS, UAL, BLDR, FAST, HON
Model 4	PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON
Model 5	PAYC PAYX DAY BR FDX UPS UAL LUV BLDR CPRT FAST HON
Model 6	PAYC PAYX DAY BR UPS UAL LUV BLDR FAST HON
Model 7	PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON
Model 8	PAYC PAYX DAY BR UPS UAL LUV BLDR FAST HON
Model 9	PAYC PAYX DAY BR UPS UAL BLDR FAST HON
Model 10	PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON

Model 11	PAYX DAY BR UPS UAL BLDR
Model 12	PAYX DAY BR UPS UAL BLDR

Models 1, 2, 3 and 9 are the same i.e., the variables obtained from forward, backward stepwise selection and one of the models from C_p criterion are the same. Similarly, models 4,7 and 10, models 6 and 8, models 11 and 12 are the same. Common variables among all models are, PAYX, DAY, BR, UPS, UAL and BLDR. These are our final independent variables without including interaction or quadratic terms.

Model	r-squared adjusted	F-statistic		Coefficient of variation	No. of Variables insignificant variables
		Value	p		
Model 1,2,3,9	0.9678	839.8	<.0001	1.27714	9 0
Model 4,7,10	0.9680	690.34	<.0001	1.27440	11 2
Model 5	0.9678	630.42	<.0001	1.27682	12 3
Model 6,8	0.9680	761.11	<.0001	1.27291	10 2
Model 11,12	0.9664	1204.75	<.0001	1.30475	6 0

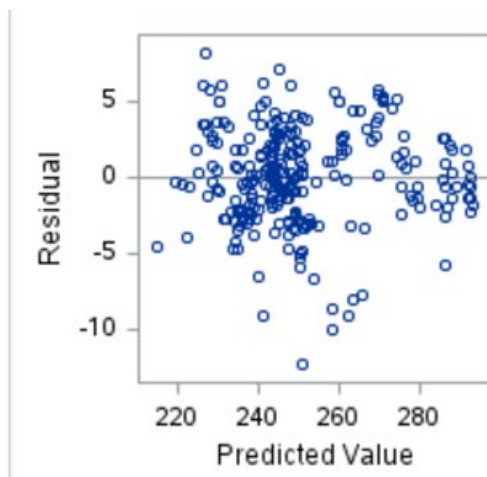
If the p-value is greater than alpha (0.05), then we fail to reject the null hypothesis (slope of the specified independent variable = 0), concluding that the particular independent variable is not significant.

Model 4,7, 10: FDX and LUV are not significant ($p = 0.5097$ and 0.0835).

Model 5: FDX, LUV, CPRT and FAST are not significant ($p = 0.4918, 0.1045, 0.7628$ and 0.0685)

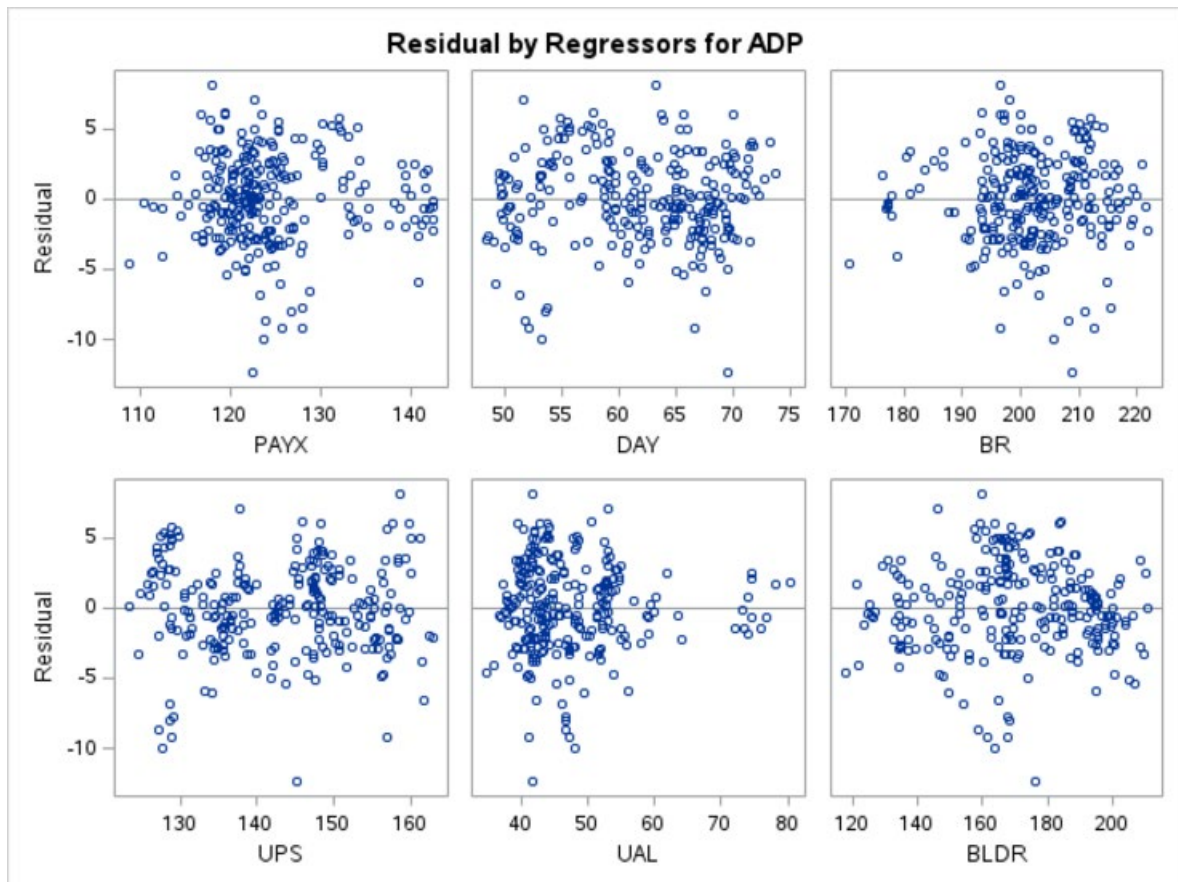
Model 6,8: LUV and FAST are not significant ($p = 0.1075$ and 0.0523)

To check for heteroscedasticity, unequal variances for difference settings of the independent variable(s), we can look at the graph between residual and predicted value:

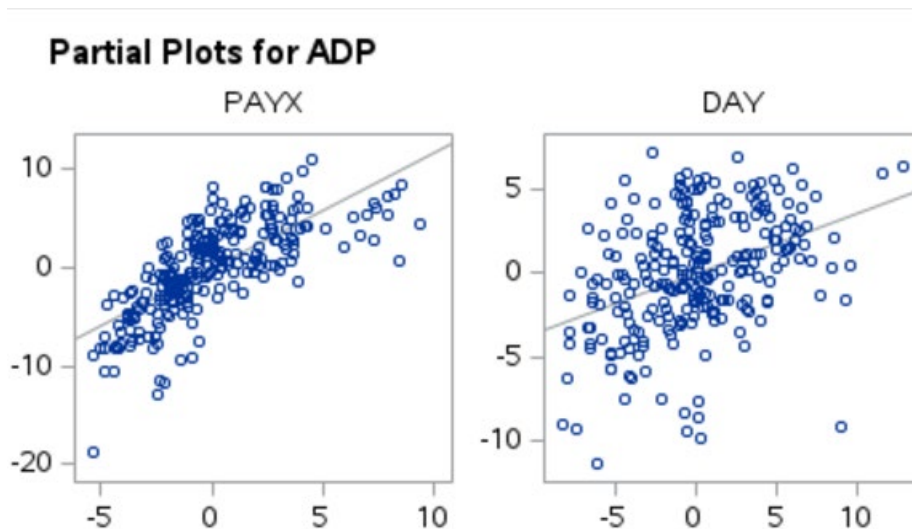


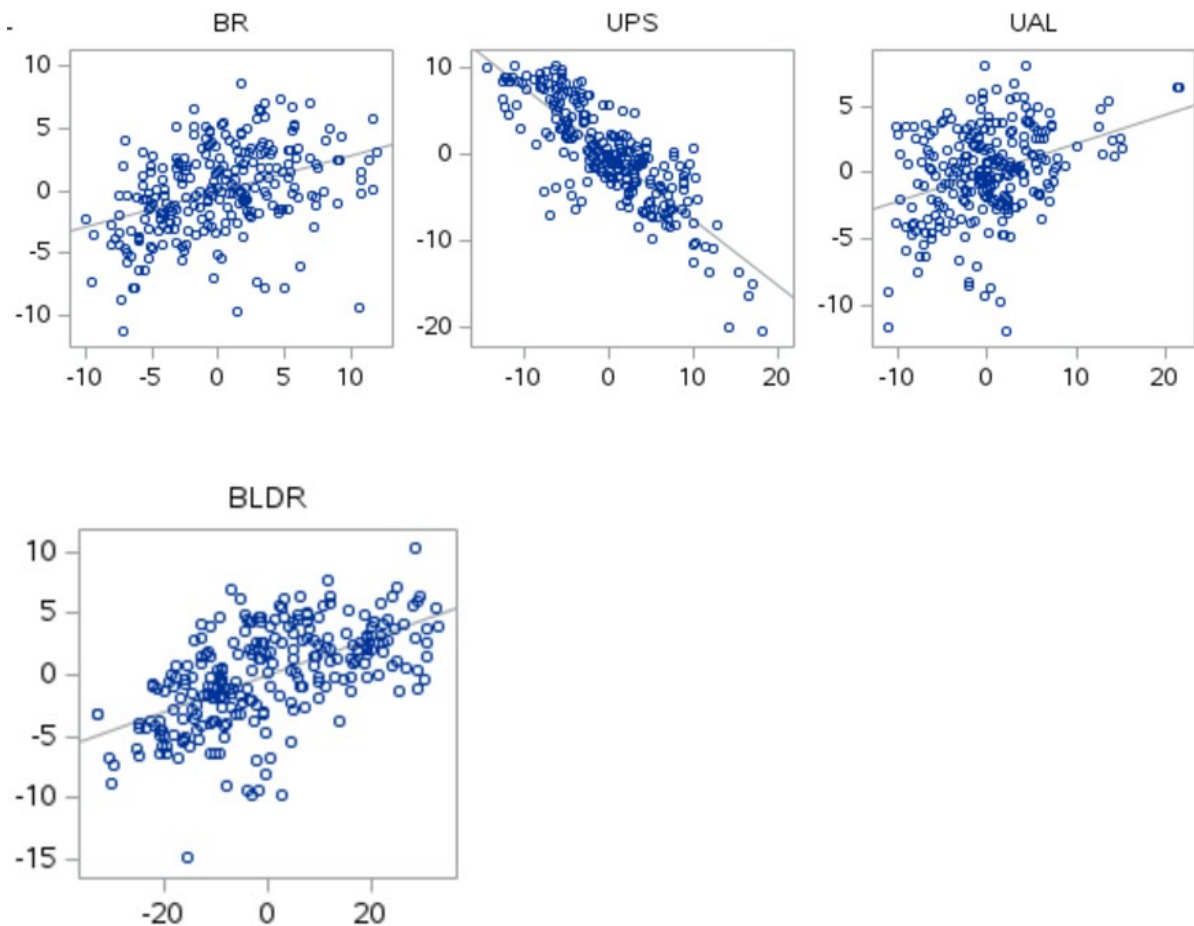
The residuals are scattered randomly around zero, indicating no major heteroscedasticity issues. However, there may be a slight clustering near the center, warranting closer inspection. To be sure of this, we can divide the data set into two parts and, run two separate regressions and look at the MSE, then calculate the F-statistic by dividing the higher MSE by lower MSE, check if the value is greater or less than the F-critical. Here, F-critical for $\alpha = 0.05$, degrees of freedom 6 and 6 is 4.284. Higher MSE is 9.03198 and lower is 5.61333. So, the F-statistic is 1.609. The F-value is less than the F critical, so we fail to reject the null hypothesis that assumption of equal variances is satisfied, concluding that there is no heteroscedasticity. So, we don't need to transform any variables.

Residual plots for PAYX, DAY, BR, UPS, UAL and BLDR:



Looking at the residual plots, UPS seems completely scattered without a pattern. DAY seems to have a curvature and all the other variables kind of look like exhibiting some type of patterns. Since there are multiple variables, it would be better to look at the partial plots of the variable to get an idea whether any of them need quadratic term.



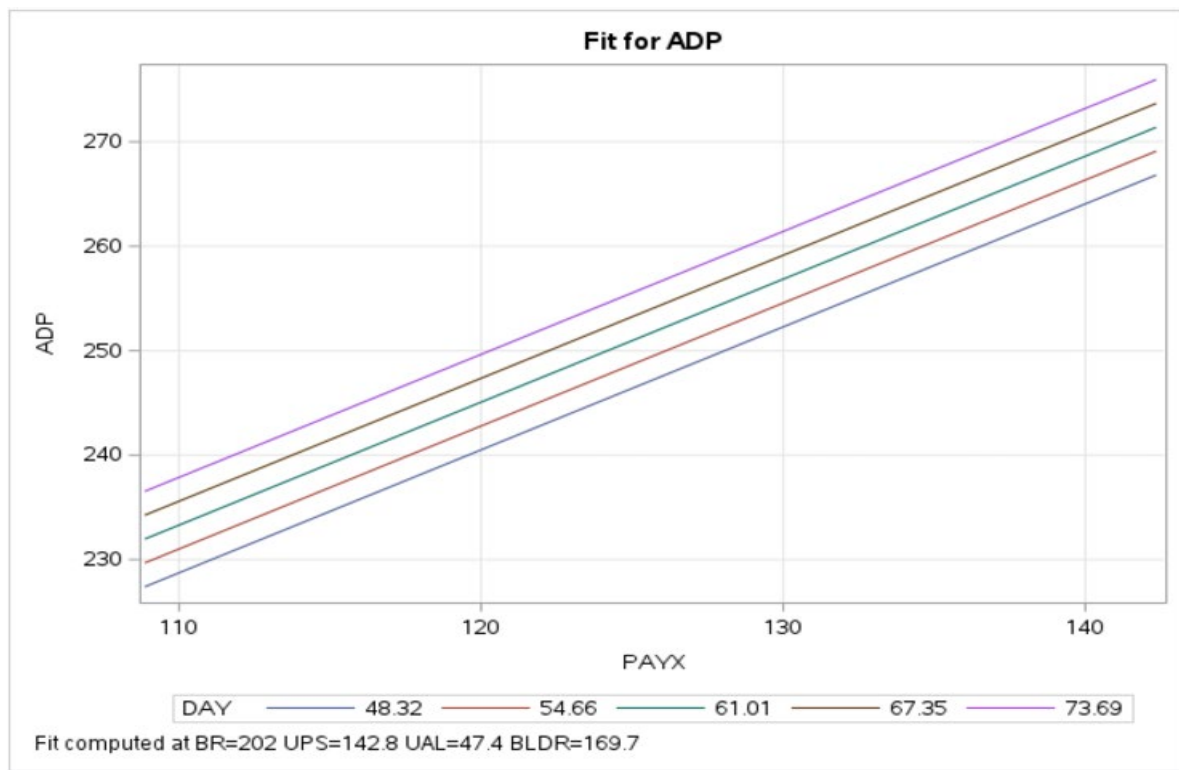


Looking at the partial residual plots, none of them seem to be exhibiting a pattern, they seem randomly scattered, except for maybe BLDR, which looks a bit like a curve. So, we can add BLDR*BLDR (quadratic term) to the model to see if the model becomes better.

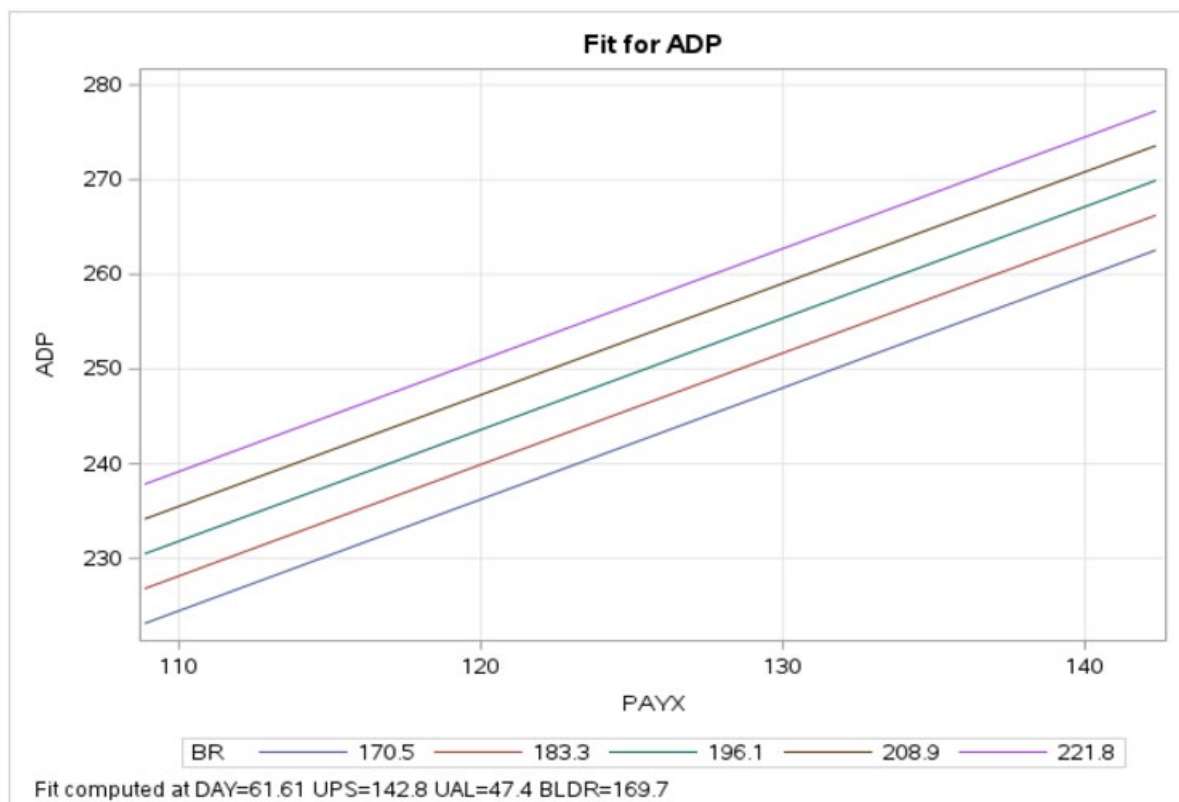
The model now contains PAYX, DAY, BR, UPS, UAL, BLDR and BLDR*BLDR. This would be our 12th model.

Now, to know if there are any interaction terms, we can look at the interaction plots. There are 6 independent variables, there will be 15 interaction terms. Here are the 15 interaction plots:

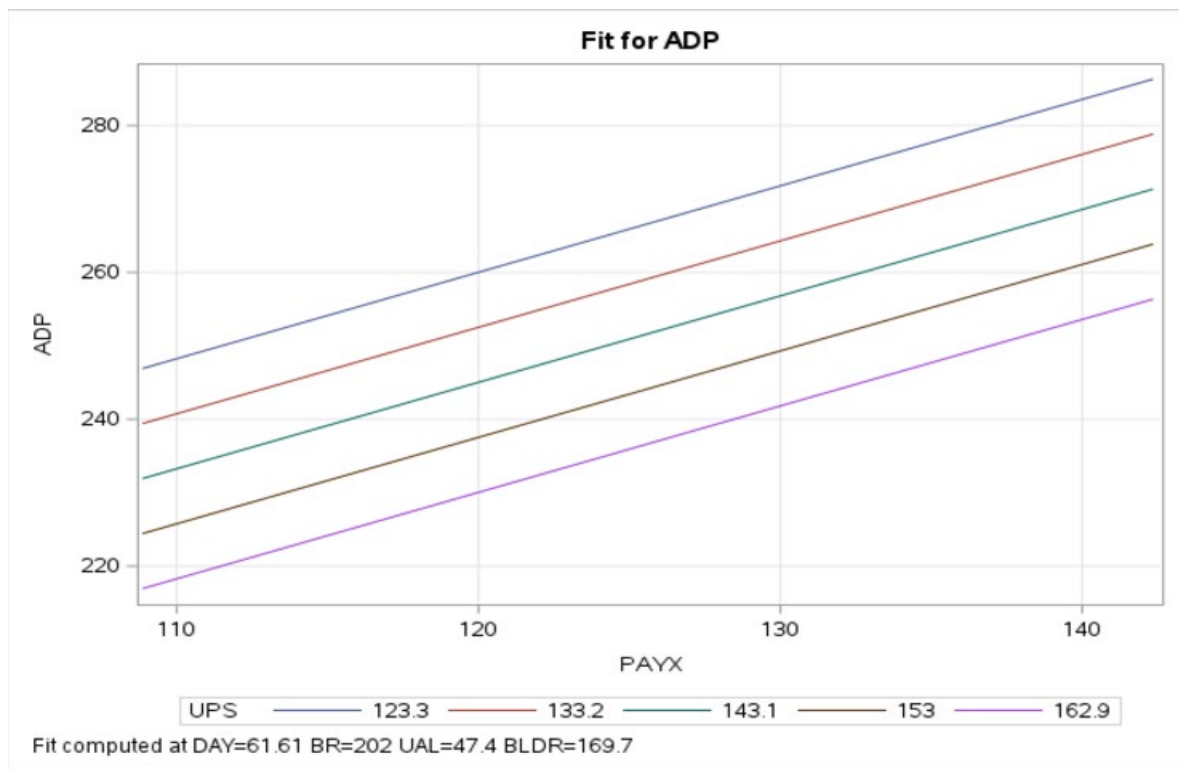
PAYX-DAY:



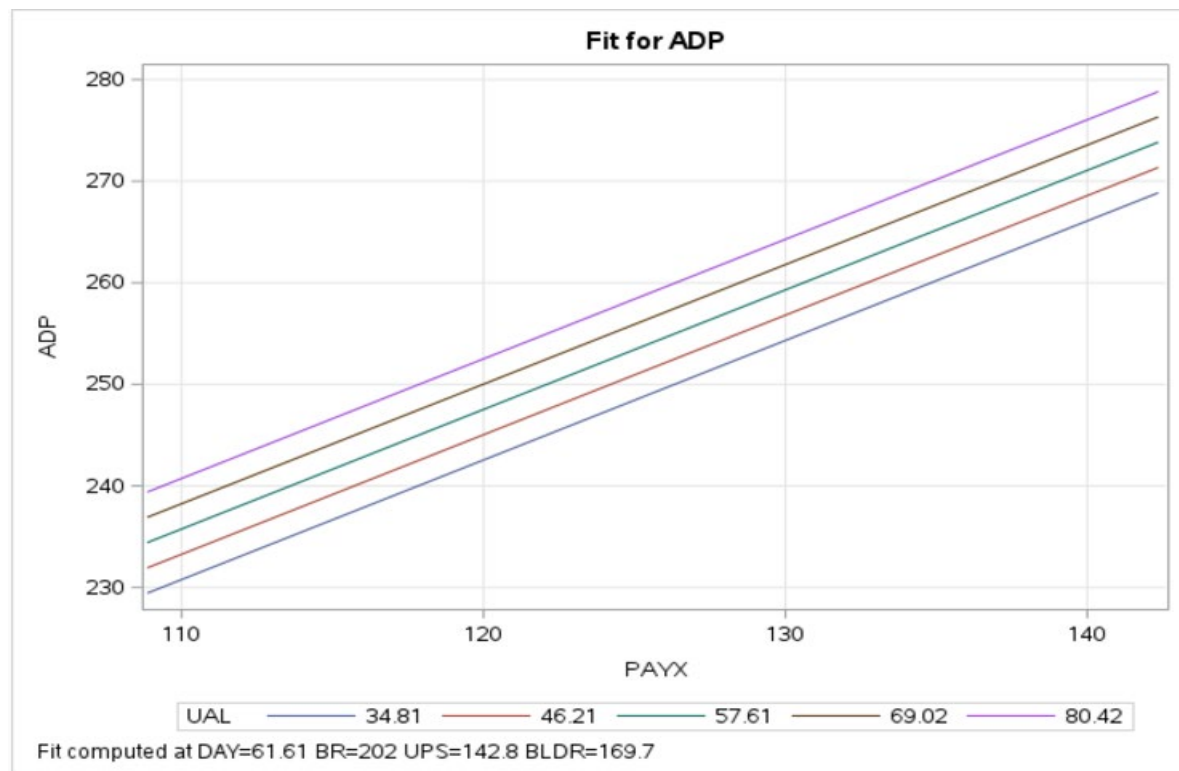
PAYX-BR:



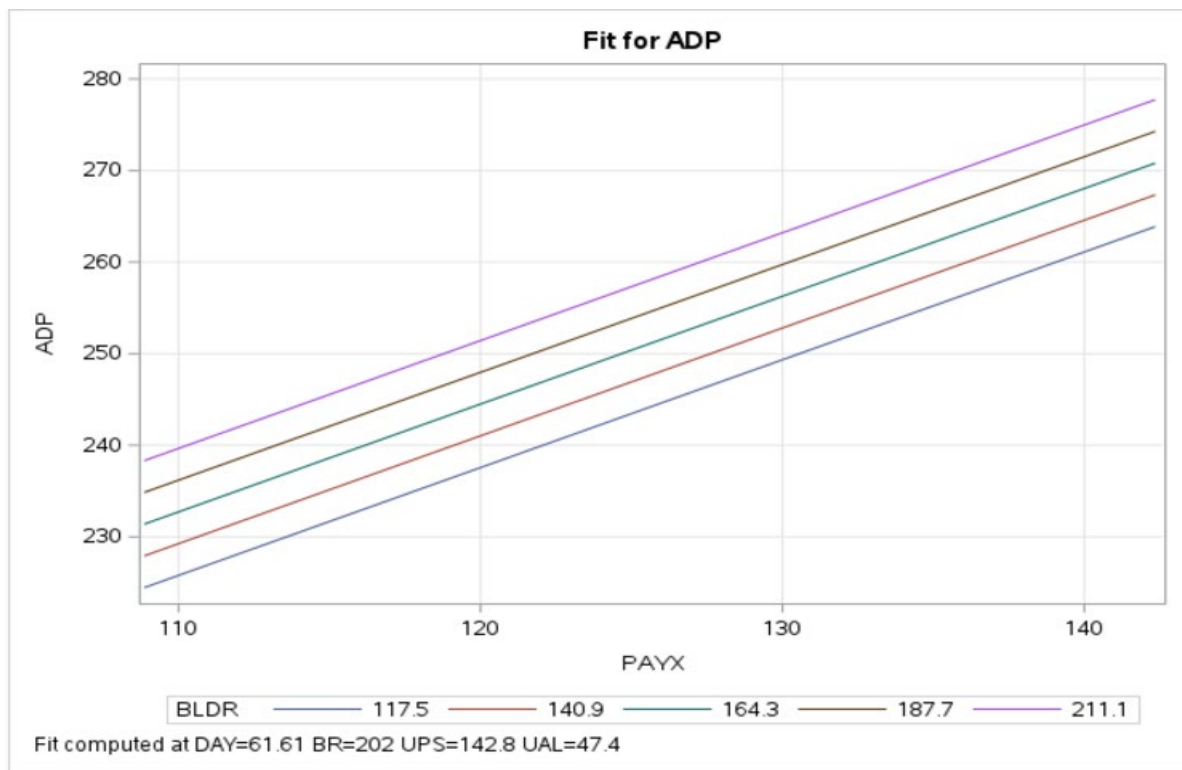
PAYX-UPS:



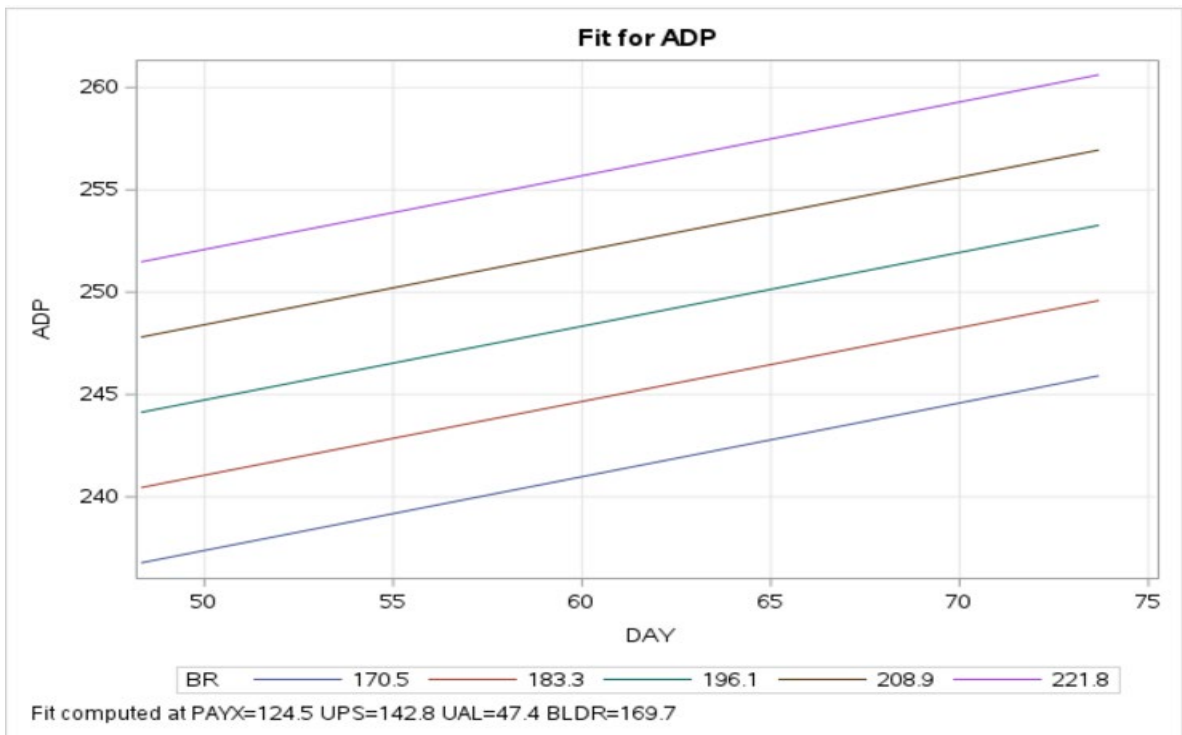
PAYX-UAL:



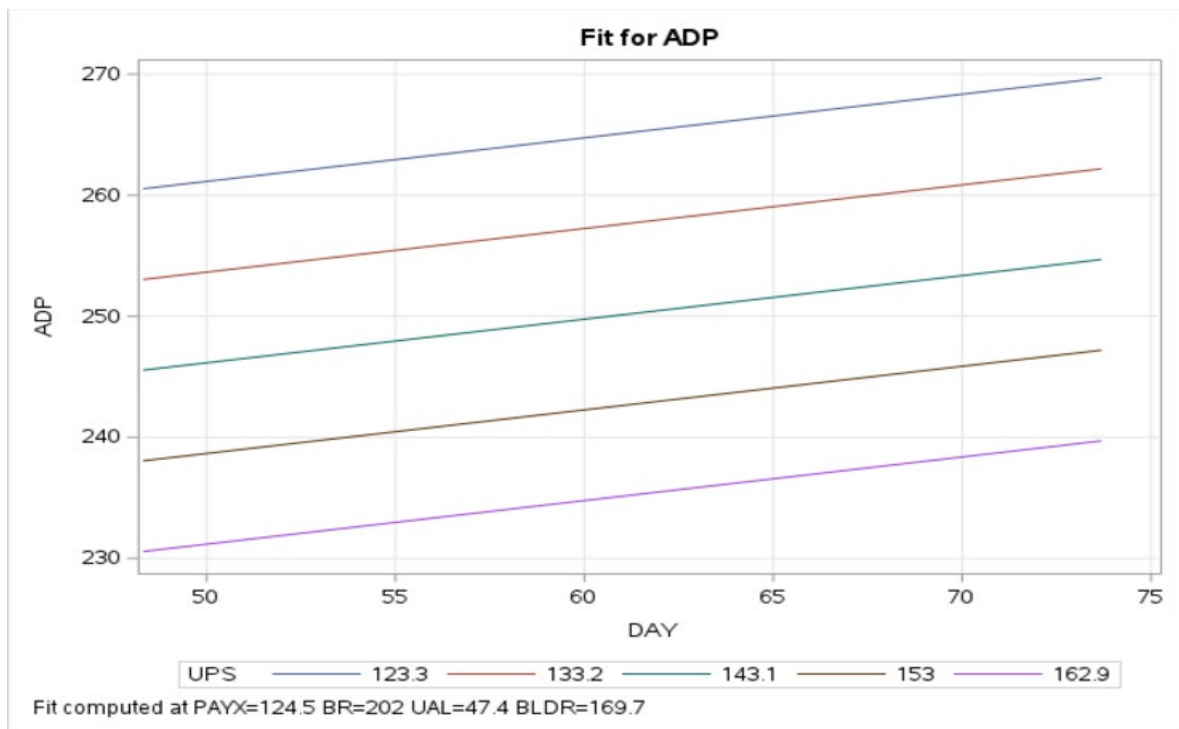
PAYX-BLDR:



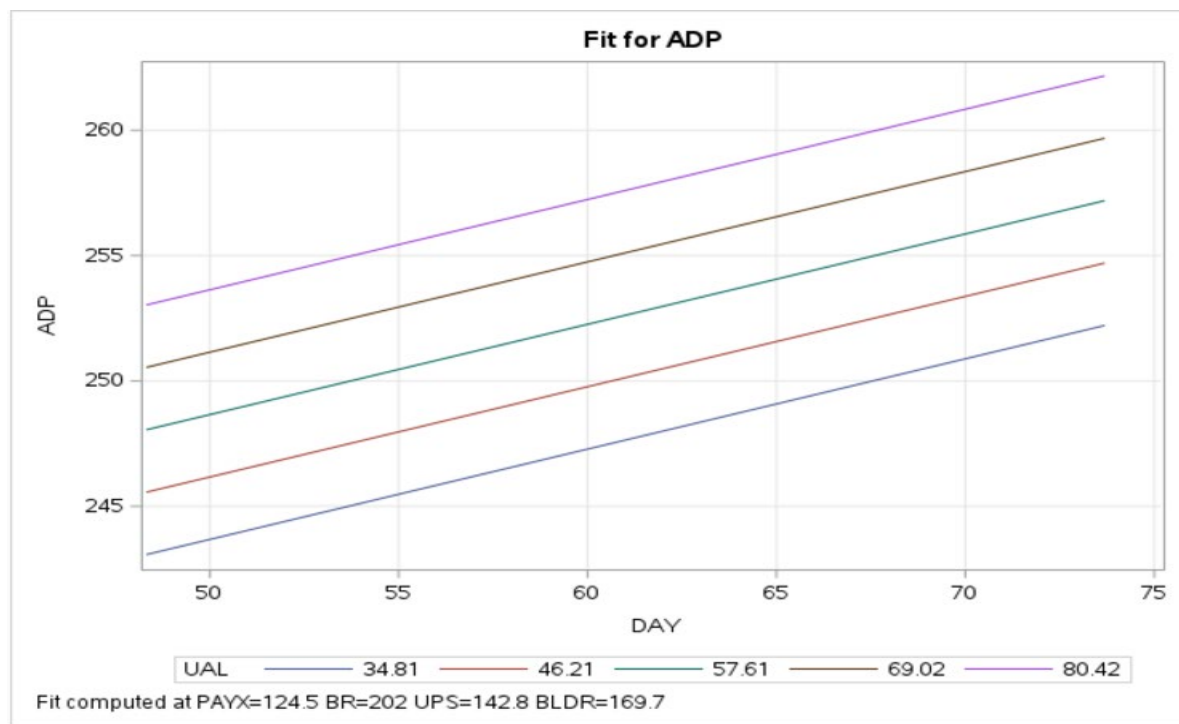
DAY-BR:



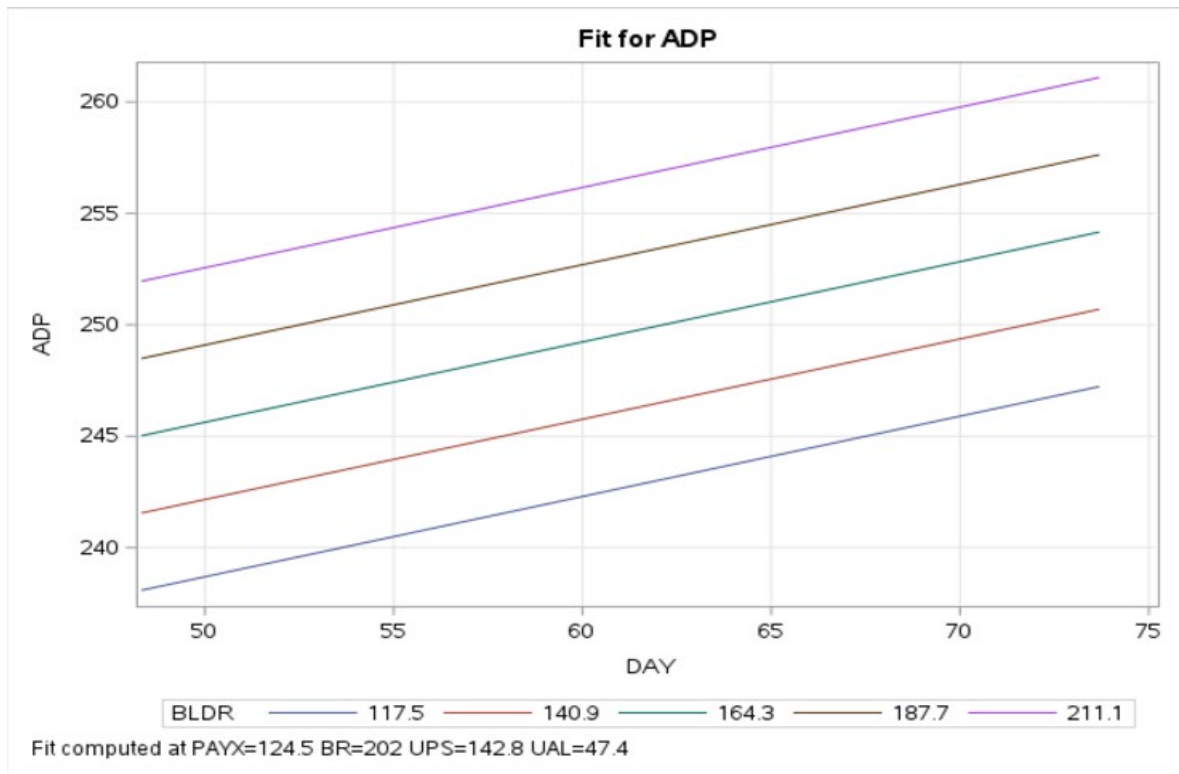
DAY-UPS:



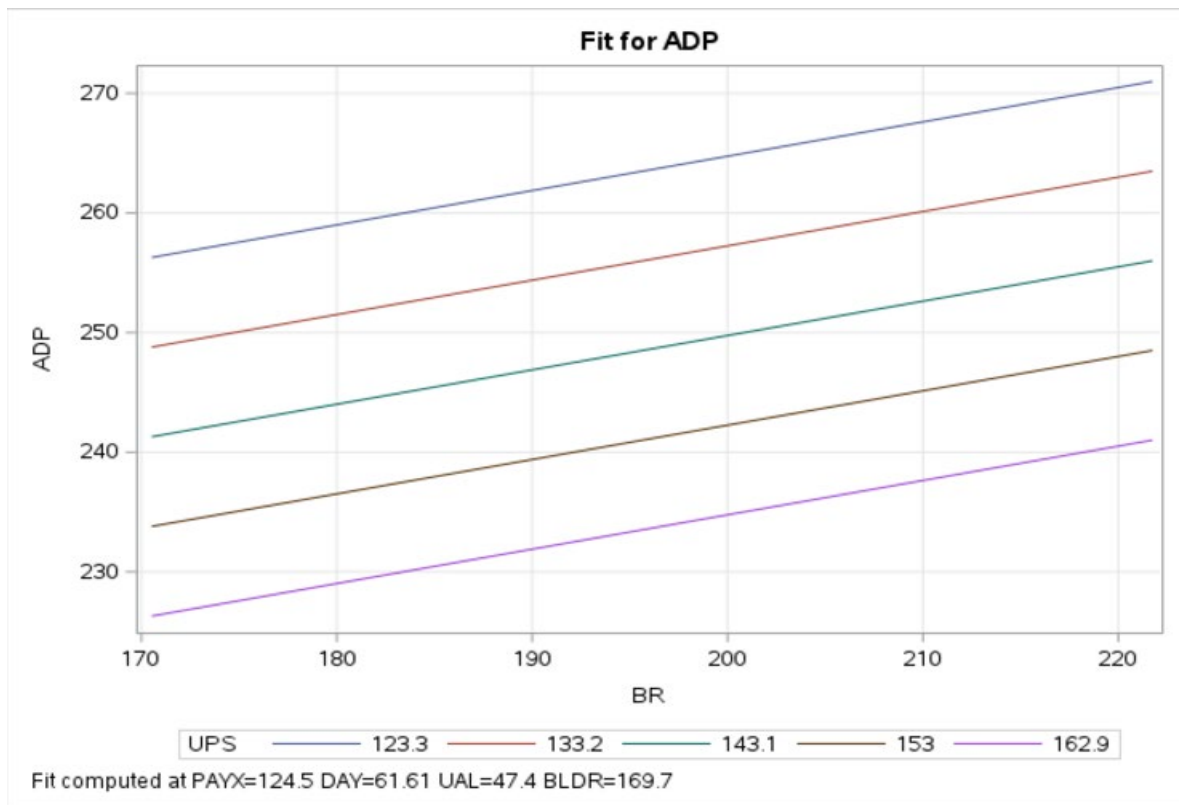
DAY-UAL:



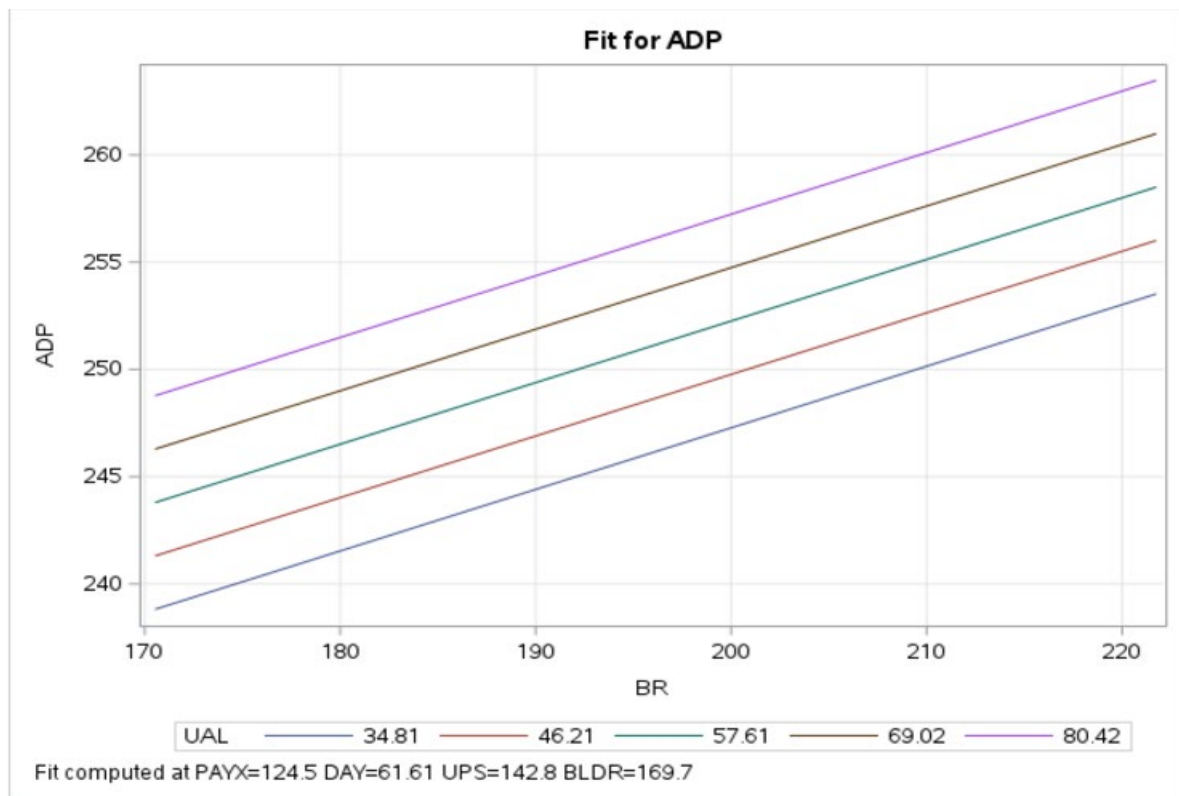
DAY-BLDR:



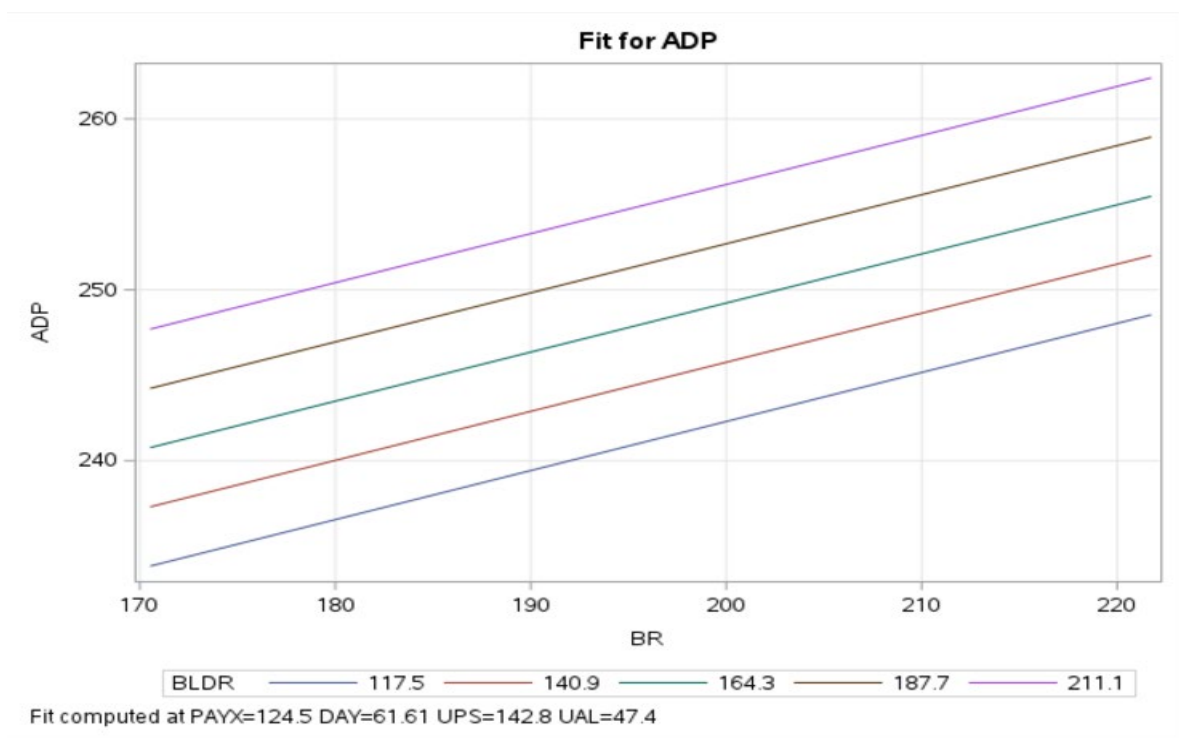
BR-UPS:



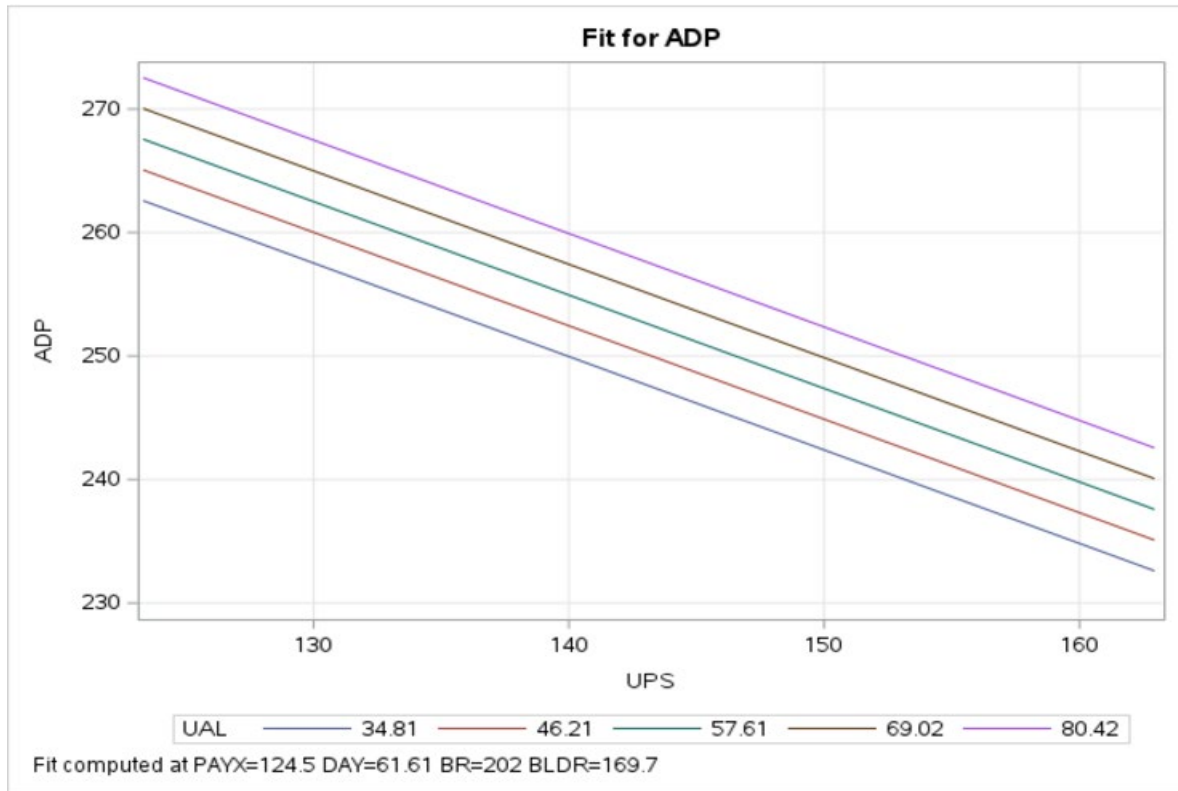
BR-UAL:



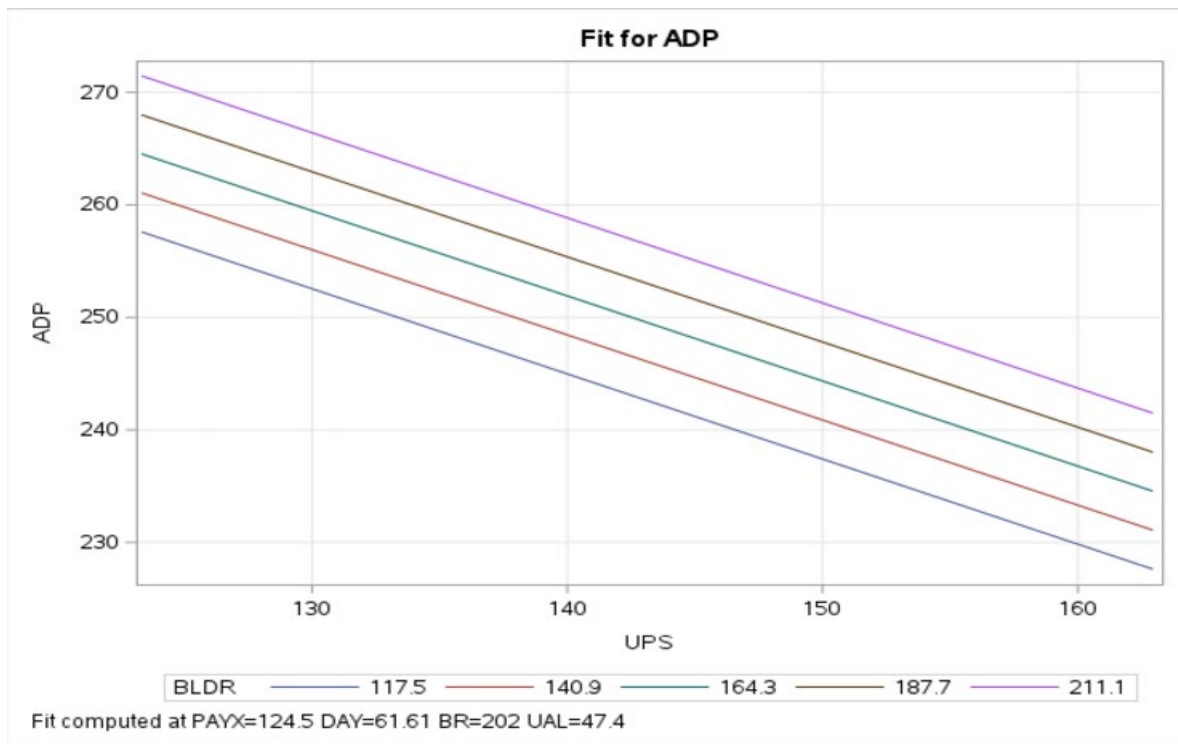
BR-BLDR:



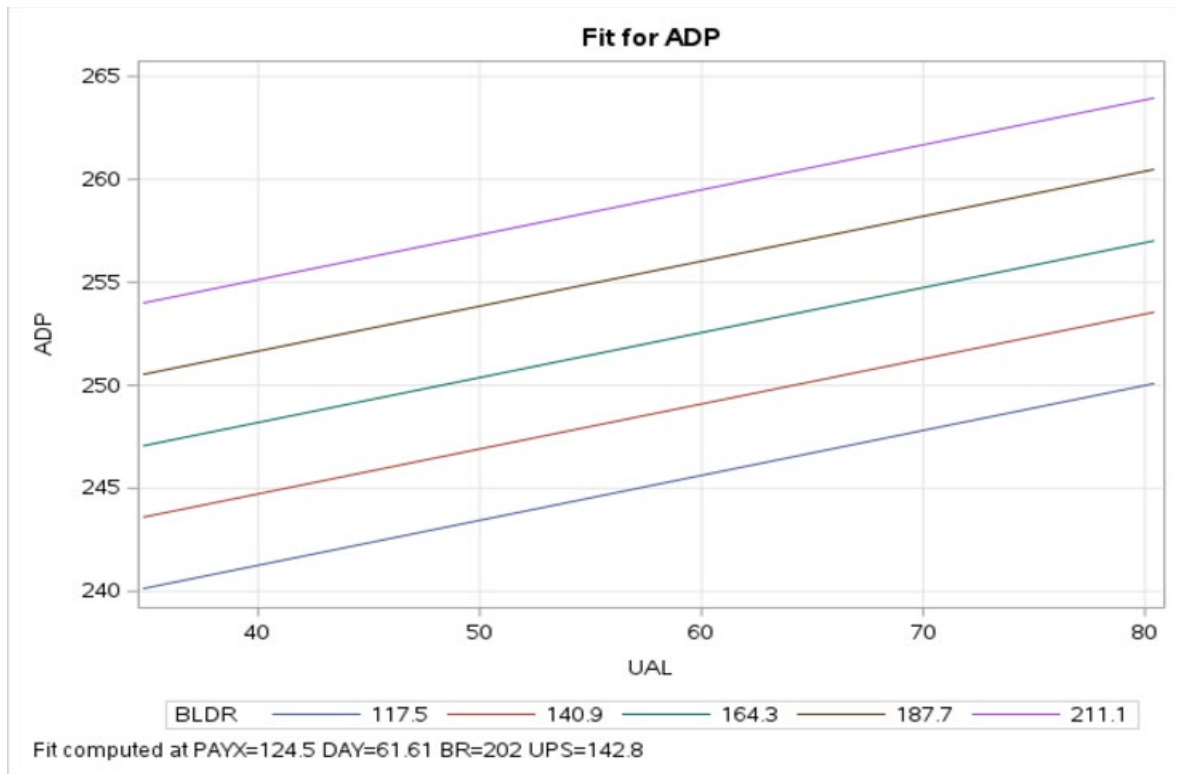
UPS-UAL:



UPS-BLDR:



UAL-BLDR:



All the graphs have parallel lines and does not seem very close to interaction. We can still try using PAYX-UAL as an interaction term as the lines are a bit closer in the graph, to see if it would make the model better.

Now we can add PAYX*UAL along with PAYX, DAY, BR, UPS, UAL and BLDR to get model 13 and add PAYX*UAL to PAYX, DAY, BR, UPS, UAL, BLDR and BLDR*BLDR, to get model 14. We can now compare the models using relevant statistics to see if which model is the best.

Model	r-squared adjusted	F-statistic	Coefficient of variation	No. of Variables insignificant variables
		Value p		
Model 11,12	0.9664	1204.75 <.0001	1.30475	6 0

Model 13	0.9677	1073.67 <.0001	1.28047	7 0
Model 14	0.9664	1031.75 <.0001	1.30539	7 2
Model 15	0.9675	936.38 <.0001	1.28259	8 2

Model 14: UAL and PAYX*UAL are not significant ($p = 0.2063$ and 0.3839)

Model 15: UAL and PAYX*UAL are not significant ($p = 0.9569$ and 0.6600)

For Model 14 and 15, the p-value of the interaction term is greater than 0.05 so, we fail to reject the null hypothesis (slope of the interaction term (PAYX*UAL) = 0) concluding that the interaction term is not statistically significant and does not add value to the model. This aligns with what we found from the interaction plot.

Compared to model 11 and 12 (the one with our final independent variables without interaction and quadratic terms), 14 (with interaction term) and 15 (with both quadratic and interaction term) doesn't seem that good since we can see that the interaction term is not significant. We can check whether model 13 (with quadratic term) is better than model 11/12 by doing an F-test for the nested models, with model 13 (with quadratic term) as the complete model and model 11/12 (the one with our final independent variables without interaction and quadratic terms) as the reduced model. The value of F-statistic is 10.38 and the p-value is 0.0014 which is less than 0.05, so we can reject the null hypothesis (slope of the quadratic term (BLDR*BLDR) = 0) concluding that the quadratic term (BLDR*BLDR) contributes information to the prediction of the daily prices of ADP.

So, our best model would be model 13 (with quadratic term). The model for predicting daily prices of ADP (Automatic data processing) is:

$$\text{ADP} = 70.01942 + 1.15194 (\text{PAYX}) + 0.39095 (\text{DAY}) + 0.24257 (\text{BR}) - 0.78862(\text{UPS}) + 0.22696 (\text{UAL}) + 0.63445 (\text{BLDR}) - 0.00143 (\text{BLDR}*\text{BLDR}).$$

We can interpret this as, if the daily prices for PAYX, DAY, BR, UPS, UAL, BLDR are zero, then the daily price for ADP would be 70.01942, if daily price of PAYX is increased by 1 unit then, the daily price of ADP increases by 1.15194 units while the daily price of DAY, BR, UPS, UAL and BLDR are kept constant. Similarly, if daily price of DAY is increased by 1 unit then, the daily price of ADP increases by 0.39095 units while the daily price of PAYX, BR, UPS, UAL and BLDR are kept constant, if daily price of BR is increased by 1 unit then, the daily price of ADP increases by 0.24257 units while the daily price of DAY, PAYX, UPS, UAL and BLDR are kept constant, if daily price of UPS is increased by 1 unit then, the daily price of ADP decreases by 0.78862 units while the daily price of DAY, BR, PAYX, UAL and BLDR are kept constant, if daily price of UAL is increased by 1 unit then, the daily price of ADP increases by 0.22696 units while the daily price of DAY, BR, UPS, PAYX and BLDR are kept constant. The coefficient of BLDR (0.63445) represents the initial rate of change of daily price of ADP with respect to BLDR when BLDR is small and all other independent variables are kept fixed i.e., for small values of BLDR, a 1 unit increase in daily price of BLDR is associated with an increase in daily price of ADP by 0.63445 units assuming BLDR*BLDR does not have a large impact and all other independent variables are kept constant.

The coefficient of the quadratic term (BLDR*BLDR) is negative indicating that the relationship between the daily prices of ADP and BLDR curves downward i.e., when daily price of BLDR increase, the daily price of ADP decrease.

The sum of squares error (SSE) is 2512.56586, standard deviation (s) and variance (s^2) are 3.20896 and 10.29740 respectively. We expect most (approximately 95%) of the

observed values of daily prices of ADP to lie within 6.41792 units of the least squares regression line.

The p-value of the F-statistic is $< .001$ which is less than 0.05, we reject the null hypothesis (slopes of PAYX, DAY, BR, UPS, UAL and BLDR = 0), concluding that the slope of at least one variable is not zero. The p-value of all the independent variables (PAYX, DAY, BR, UPS, UAL and BLDR) is less than 0.05, we reject the null hypotheses (slopes of PAYX, DAY, BR, UPS, UAL and BLDR = 0, separately) concluding that there is a relationship between daily prices of ADP and daily prices of PAYX, DAY, BR, UPS, UAL and BLDR. The p-value of the quadratic term (BLDR*BLDR) is less than 0.05, we reject the null hypothesis (slope of BLDR*BLDR = 0) concluding that the quadratic term (BLDR*BLDR) contributes information to the prediction of the daily prices of ADP. The value of r-squared adjusted is 0.9677, which means 96.77% of the variance in daily prices of ADP can be explained by the model.

Final Conclusions and Next Steps

In conclusion, the quadratic model incorporating PAYX, DAY, BR, UPS, UAL, BLDR, and the quadratic term $BLDR * BLDR$ emerged as the most effective model for predicting ADP's daily stock prices. This model accounted for 96.77% of the variance in ADP's daily stock prices, suggesting a high degree of accuracy in explaining the relationship between the predictors and ADP's stock performance. The model reveals that when the daily prices of PAYX, DAY, BR, UPS, UAL, and BLDR are at zero, the daily stock price of ADP would be 70.01942. Additionally, the analysis highlights an inverse relationship between ADP's daily stock price and BLDR's daily price—indicating that as BLDR's daily price increases, ADP's stock price decreases. This insight points to a significant interaction between ADP and BLDR, where changes in one stock price are reflected in the other in opposite directions.

Moving forward, I plan to explore using all the independent variables available in the dataset to determine if a more comprehensive model can improve predictions. This will allow us to better understand the contributions of other potentially influential factors. I also intend to experiment with higher-order terms, including cubic or quartic terms, in addition to quadratic and interaction terms, to capture more complex relationships in the data that may be missed by a simpler model. Additionally, leveraging advanced modeling techniques, such as regularization methods (LASSO, Ridge regression) or machine learning algorithms (e.g., Random Forest, Gradient Boosting), could further enhance the robustness and accuracy of the model. By incorporating a broader set of interaction and higher-order terms, we may uncover additional nuances in the data, leading to a more accurate and predictive model for ADP's daily stock prices. These next steps aim to refine the model's predictive power and ensure its applicability in real-world forecasting scenarios.

Appendix

Code Used:

```
proc corr data = lab3;
```

```
Var adp payc payx day br efx fdx ups chrw dal ual luv carr jci bldr ctas cprr ldos fast uri  
hon mmm uber;
```

```
run;
```

```
proc reg data = lab3;
```

```
model adp = payc payx day br efx fdx ups chrw dal ual luv carr jci bldr ctas cprr ldos fast  
uri hon mmm uber/selection = stepwise slentry = 0.05 slstay = 0.05;
```

```
run;
```

Fig 1:

Summary of Stepwise Selection									
Step	Variable Entered	Variable Removed	Label	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	PAYX		PAYX	1	0.8477	0.8477	940.346	1391.79	<.0001
2	UPS		UPS	2	0.0631	0.9108	449.889	176.21	<.0001
3	BLDR		BLDR	3	0.0467	0.9575	87.4807	272.64	<.0001
4	DAY		DAY	4	0.0017	0.9592	76.1250	10.37	0.0015
5	BR		BR	5	0.0035	0.9627	51.1659	22.78	<.0001
6	UAL		UAL	6	0.0045	0.9672	17.8358	33.83	<.0001
7	FAST		FAST	7	0.0005	0.9677	15.7009	4.01	0.0464
8	PAYC		PAYC	8	0.0006	0.9684	12.7814	4.84	0.0287
9	HON		HON	9	0.0006	0.9690	10.1194	4.66	0.0319

```
proc reg data = lab3;
```

```
model adp = payc payx day br efx fdx ups chrw dal ual luv carr jci bldr ctas cprr ldos fast  
uri hon mmm uber/selection = forward slentry = 0.05 slstay = 0.05;
```

```
run;
```

Fig 2:

Summary of Forward Selection								
Step	Variable Entered	Label	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	PAYX	PAYX	1	0.8477	0.8477	940.346	1391.79	<.0001
2	UPS	UPS	2	0.0631	0.9108	449.889	176.21	<.0001
3	BLDR	BLDR	3	0.0467	0.9575	87.4807	272.64	<.0001
4	DAY	DAY	4	0.0017	0.9592	76.1250	10.37	0.0015
5	BR	BR	5	0.0035	0.9627	51.1659	22.78	<.0001
6	UAL	UAL	6	0.0045	0.9672	17.8358	33.83	<.0001
7	FAST	FAST	7	0.0005	0.9677	15.7009	4.01	0.0464
8	PAYC	PAYC	8	0.0006	0.9684	12.7814	4.84	0.0287
9	HON	HON	9	0.0006	0.9690	10.1194	4.66	0.0319

```
proc reg data = lab3;
```

```
    model adp = payc payx day br efx fdx ups chrw dal ual luv carr jci bldr ctas cprr ldos fast  
    uri hon mmm uber/selection = backward slentry = 0.05 slstay = 0.05;
```

```
run;
```

Fig 3:

Summary of Backward Elimination								
Step	Variable Removed	Label	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	CPRT	CPRT	11	0.0000	0.9694	11.0913	0.09	0.7628
2	FDX	FDX	10	0.0001	0.9693	9.5257	0.44	0.5097
3	LUV	LUV	9	0.0003	0.9690	10.1194	2.61	0.1075

Code Used:

```
proc print data = lab3 (obs = 10); run;
```

```
proc corr data = lab3;
```

```
    Var adp payc payx day br efx fdx ups chrw dal ual luv carr jci bldr ctas cprr ldos fast uri  
    hon mmm uber;
```

```
run;
```

```
proc reg data = lab3;
```

```
    model adp = payc payx day br fdx ups ual luv bldr cprr fast hon/vif;
```

```
run;
```

```
proc reg data = lab3;
```

```
    model adp = payc payx day br fdx ups ual luv bldr cprr fast hon/SELECTION =  
    STEPWISE slentry = 0.05;
```

```
    run;
```

```
proc reg data = lab3;
```

```
    model adp = payc ups bldr day br ual fast payx hon;
```

```
    run;
```

```
/*forward*/
```

```
proc reg data = lab3;
```

```
    model adp = payc payx day br fdx ups ual luv bldr cprr fast hon/selection = forward  
    slentry = 0.05 slstay = 0.05;
```

```
    run;
```

```
/*backward*/
```

```
proc reg data = lab3;
```

```
    model adp = payc payx day br fdx ups ual luv bldr cprr fast hon/selection = backward  
    slstay = 0.05;
```

```
    run;
```

```
proc reg data = lab3;
```

```
    model adp = PAYC PAYX DAY BR UPS UAL BLDR FAST HON;
```

```
    run;
```

```
/*cp*/
```

```
proc reg data = lab3;
```

```
    model adp = payc payx day br fdx ups ual luv bldr cprr fast hon/selection = cp slentry =  
    0.05 slstay = 0.05;
```

```
    run;
```

```
proc reg data = lab3;
```



```
model adp = PAYC PAYX DAY BR UPS UAL LUV BLDR FAST HON ;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYC PAYX DAY BR UPS UAL BLDR FAST HON;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON;  
run;
```

```
/*adjrsq*/  
proc reg data = lab3;  
    model adp = payc payx day br fdx ups ual luv bldr cprt fast hon/selection = adjrsq slentry  
= 0.05 slstay = 0.05;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYC PAYX DAY BR UPS UAL LUV BLDR FAST HON;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON;  
run;
```

```
/*rsquare*/  
proc reg data = lab3;  
    model adp = payc payx day br fdx ups ual luv bldr cprt fast hon/selection = rsqaure  
slentry = 0.05 slstay = 0.05;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYC PAYX DAY BR FDX UPS UAL LUV BLDR FAST HON ;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYC PAYX DAY BR FDX UPS UAL LUV BLDR CPRT FAST HON;  
run;
```

```
/*press*/
```

```
proc glmselect data = lab3;  
    model adp = payc payx day br fdx ups ual luv bldr cp rt fast hon/selection =  
stepwise(choose = press) slentry = 0.05 slstay = 0.05;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYX DAY BR UPS UAL BLDR;  
run;
```

```
proc glmselect data = lab3 plots = criterionpanel;  
    model adp = payc payx day br efx fdx ups chrw dal ual luv carr jci bldr ctas cp rt ldos fast  
uri hon mmm uber/ selection = stepwise stats = all;  
run;
```

```
proc glmselect data = lab3 plots = criterionpanel;  
    model adp = PAYX DAY BR UPS UAL BLDR FAST PAYC HON/ selection = stepwise  
stats = all;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYX DAY BR UPS UAL BLDR/r;  
run;
```

```
proc reg data = lab3;  
    model adp = PAYX DAY BR UPS UAL BLDR/partial;  
run;
```

```
proc freq data = lab3;  
    tables adp;  
run;
```

```
data lab3_2 lab3_3;  
    set lab3;  
    if adp <= 246.309998 then output lab3_2;  
    else output lab3_3;
```

```
proc reg data = lab3_2;  
    model adp = PAYX DAY BR UPS UAL BLDR;  
run;
```

```
proc reg data = lab3_3;  
    model adp = PAYX DAY BR UPS UAL BLDR;  
run;
```

```
proc GLM data = lab3;  
    model adp = PAYX DAY BR UPS UAL BLDR/ solution; /* order does not matter */  
    store GLMMODEL; /* name of the stored model */
```

```
proc PLM restore = GLMMODEL noinfo; /* no info - no other data, only interaction plot */  
    effectplot slicefit(x = PAYX sliceby = ual) /*- default is five number summary, u can  
choose how many and whatever u want */;  
run;
```

```
data lab3_4;  
    set lab3;  
    s1 = payx*payx;  
    s2 = day*day;  
    s3 = br*br;  
    s4 = ups*ups;
```

```

s5 = ual*ual;
s6 = bldr*bldr;
int1 = PAYX*UAL;
run;

```

```

proc reg data = lab3_4;
    model adp = PAYX DAY BR UPS UAL BLDR s6;
    test s6;
run;

```

```

proc reg data = lab3_4;
    model adp = PAYX DAY BR UPS UAL BLDR int1;
run;

```

```

proc reg data = lab3_4;
    model adp = PAYX DAY BR UPS UAL BLDR s6 int1;
run;

```

Final Model:

